



SKG14 IC8 Pacing and Emerging Technologies for ICM Engineers, L2

18.0 – What are the Machine Learning (ML) & Deep Learning (DL)

July 2025

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Learning Outcome

At the end of the session, you will be able to:

- Explain the basic fundamentals of data statistics and analytics.
- Describe the common terminologies of AI and its subsects, ML and DL.
- Apply data analytics, AI and their subsets usage in various ICM applications.
- Describe the characteristics of computing requirements most suitable for AI.

Pre-Quiz

Interactive Learning

18.1 Introduction to ML

- Features of Machine Learning
- Types of Machine Learning
- Differences between Machine Learning Types
- Machine Learning Pipeline
- Machine Learning Framework
- Advantages and Disadvantages of ML

Introduction to ML

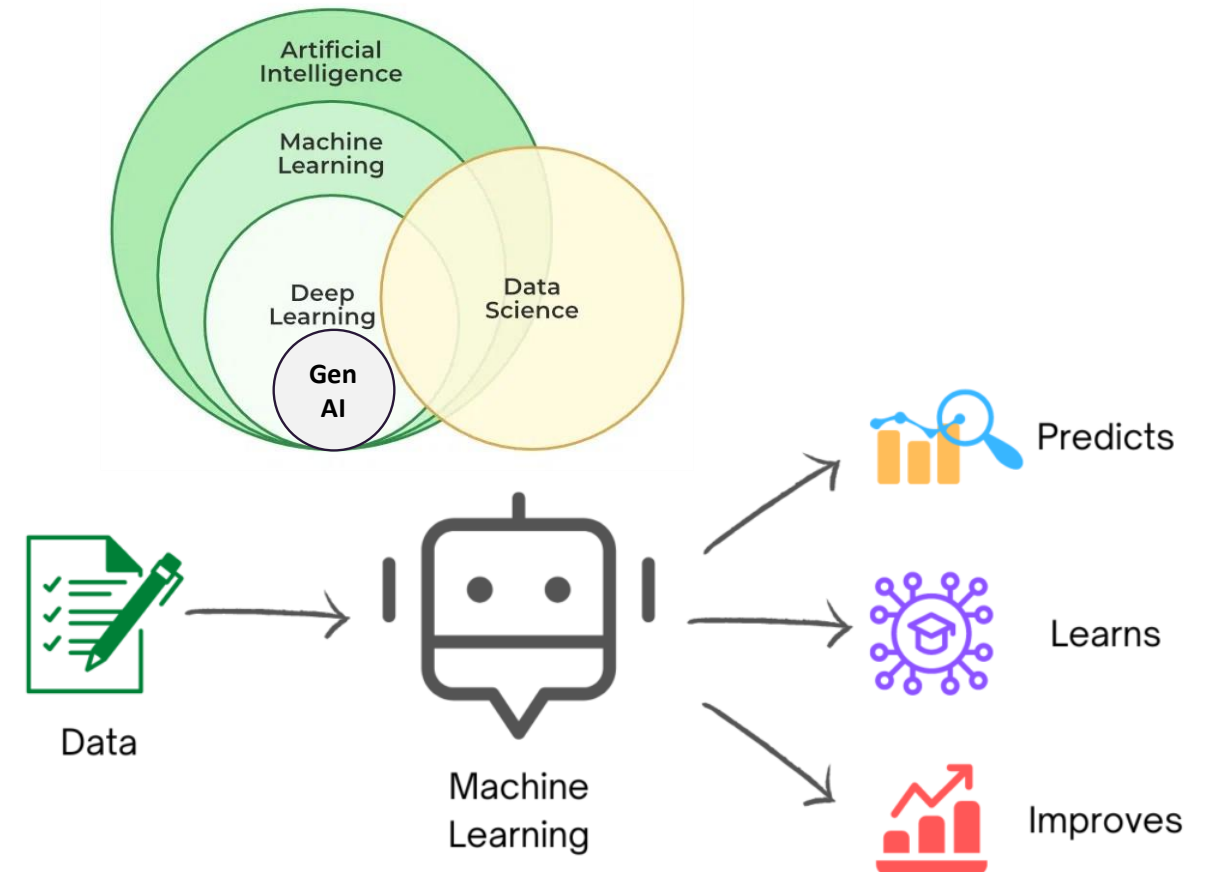
Machine Learning (ML) is a subset of Artificial Intelligence (AI) focused on developing systems that can learn and adapt from data without explicit programming.

Learning from Data:

- Machine Learning involves feeding data into algorithms to identify patterns and make predictions on new data.
- Machine learning can learn and train from data and solve/predict complex solutions, which traditional programming cannot do.
- It enables us to make better decisions and solve complex business problems in optimized time.

Improving Over Time:

- ML algorithms improve their performance over time as they are exposed to more data.
- They can adapt to new information and make more accurate predictions.



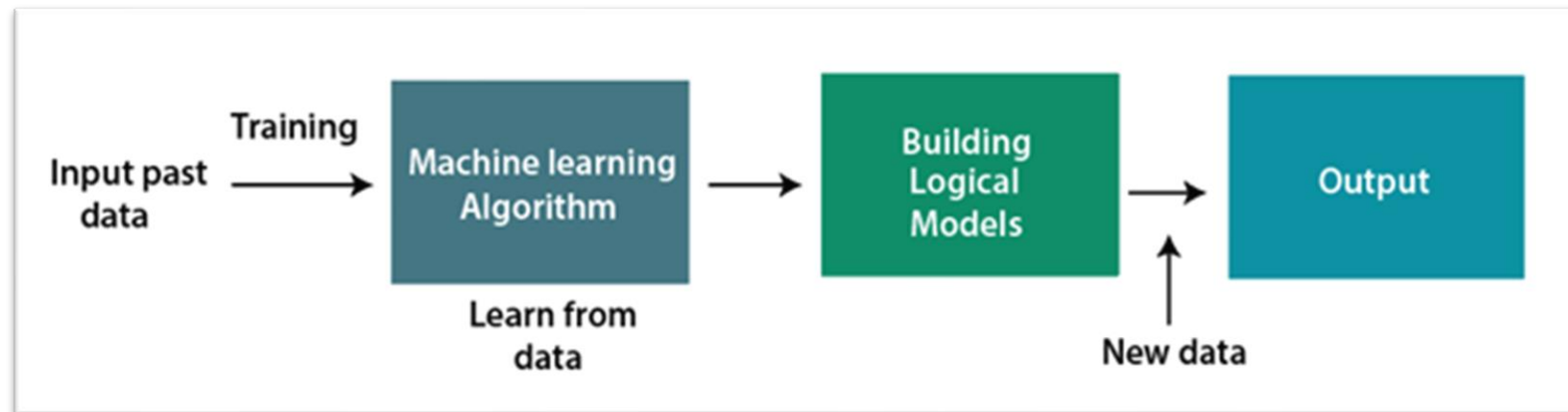
Source: <https://doi.org/10.1007/978-3-319-20010-1>

18.1 Introduction to ML

- Features of Machine Learning
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Features of Machine Learning

- To solve a complex problem for prediction:
- Conventional → write a code to solve
- With ML → feed the data to ML algorithms, it builds a logic/mathematical model, & predicts the output



- Data is used to identify various patterns in a given dataset.
- Learns from past data and improves automatically.
- Can deal with a huge amount of data.

Finding hidden patterns and extracting useful information from data

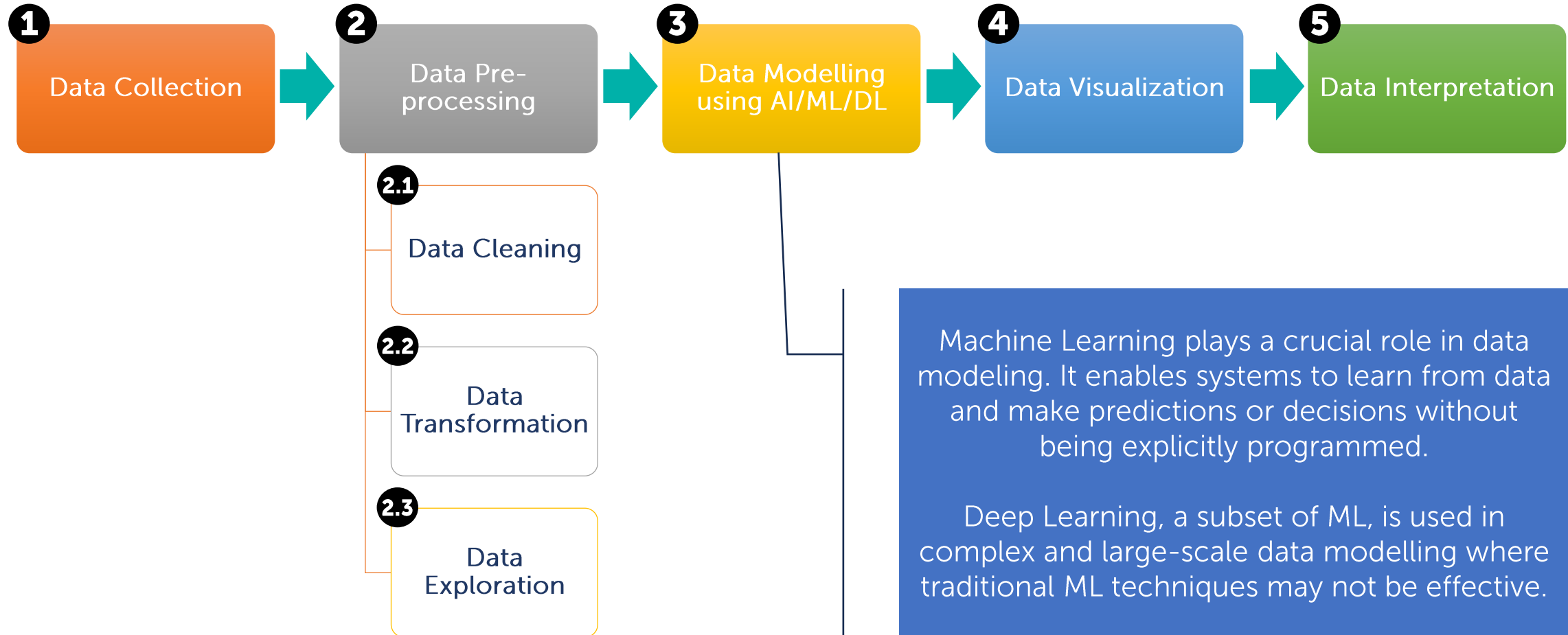
Source: <https://doi.org/10.1007/978-3-319-20010-1>

18.1 Introduction to ML

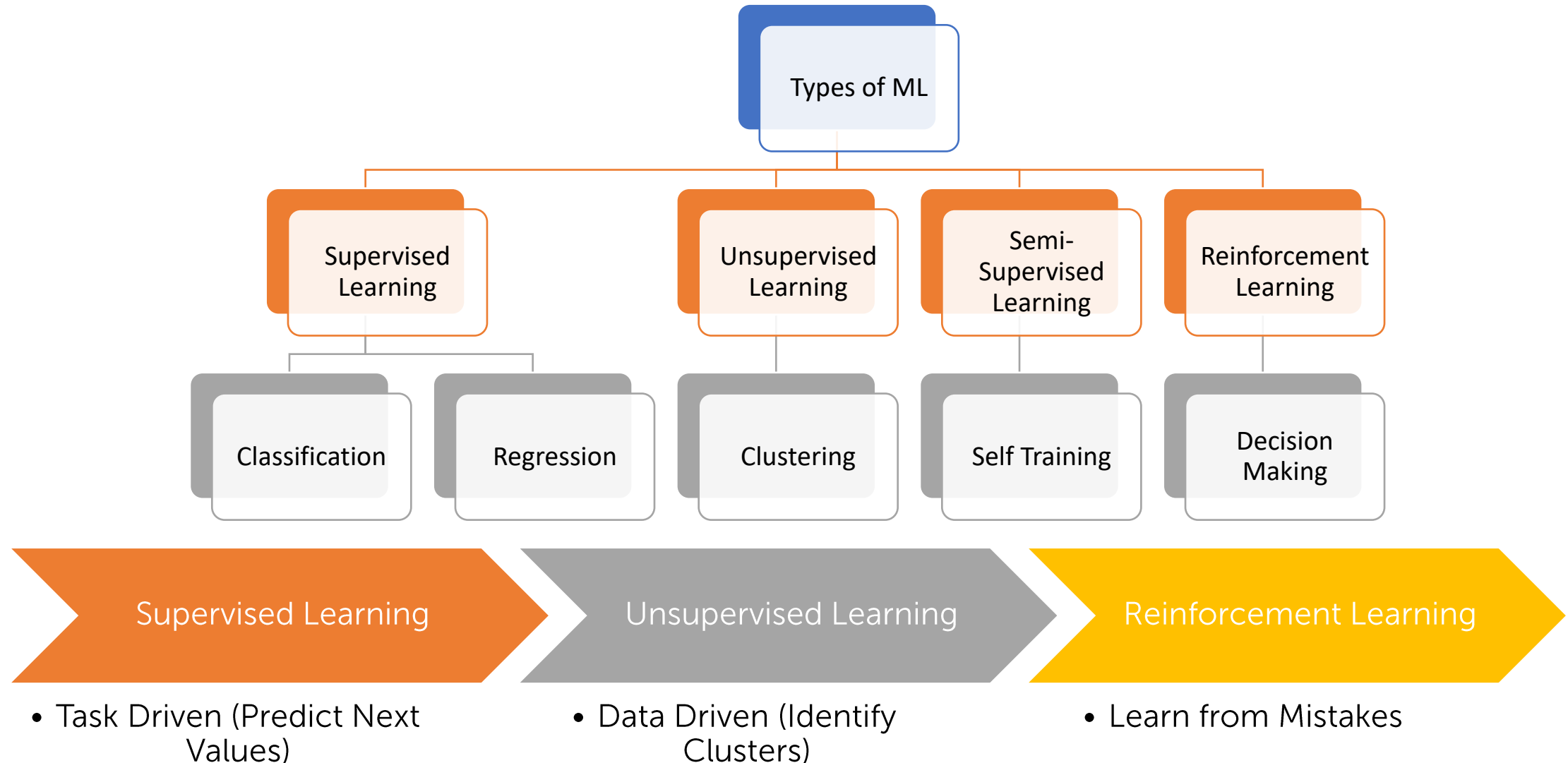
- Features of Machine Learning
- **Types of Machine Learning**
- Differences between Machine Learning Types
- Machine Learning Pipeline
- Machine Learning Framework
- Advantages and Disadvantages of ML

Data Analysis Process

Data need to analyzed by following this process to achieve some insights such as characteristics and patterns



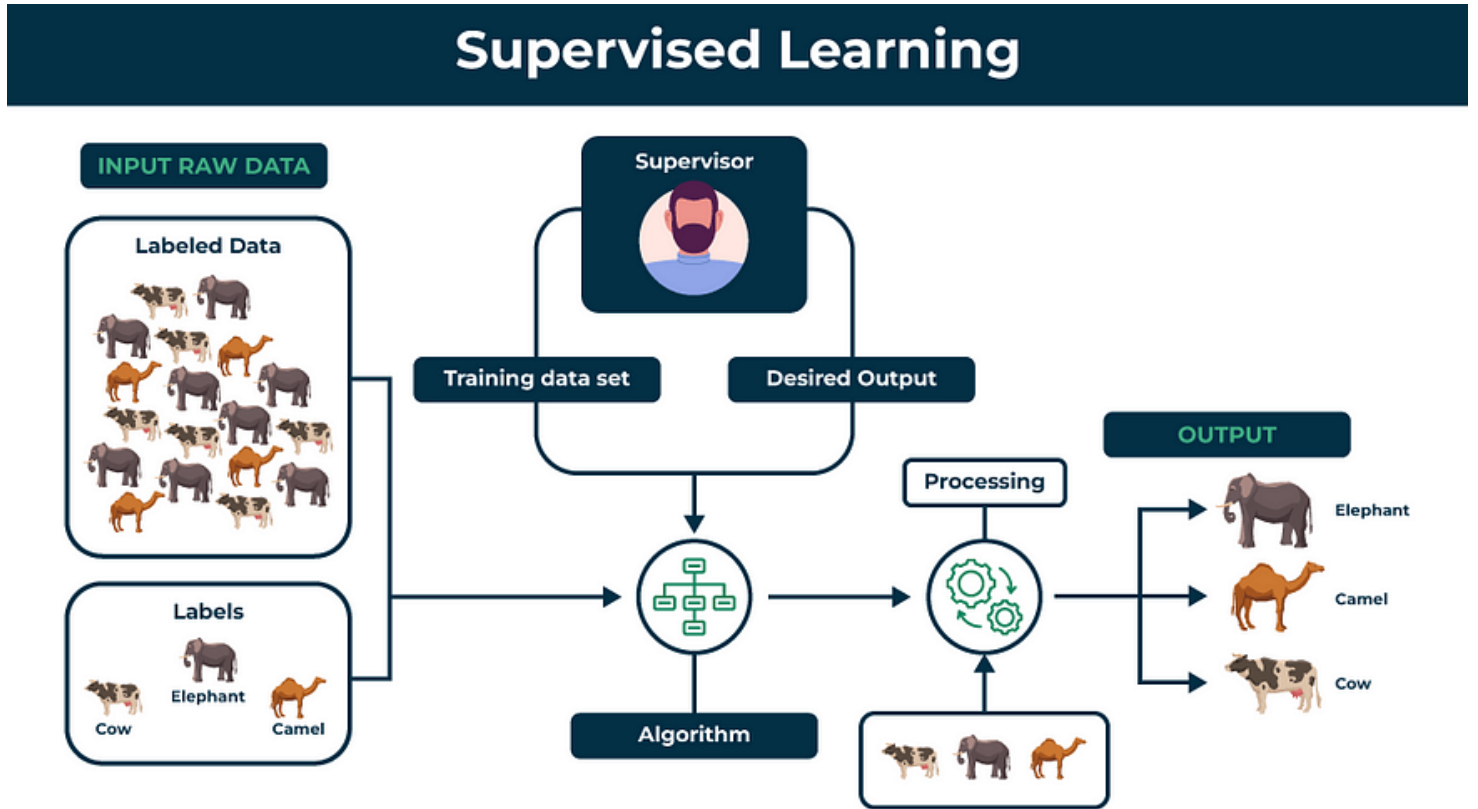
Types of Machine Learning



Machine Learning

- Supervised Learning

- **Definition:** The algorithm learns from a “Labelled Dataset” where each input is associated with a known output.
- **Goal:** To predict the correct output for new, unseen inputs.
- **Techniques:**
 - **Regression:** Predicting a continuous value (e.g., forecasting stock prices).
 - **Classification:** Predicting a category (e.g., email spam detection).
- **Applications:**
 - Predicting house prices based on size, location, and other features.
 - Diagnosing diseases based on symptoms.

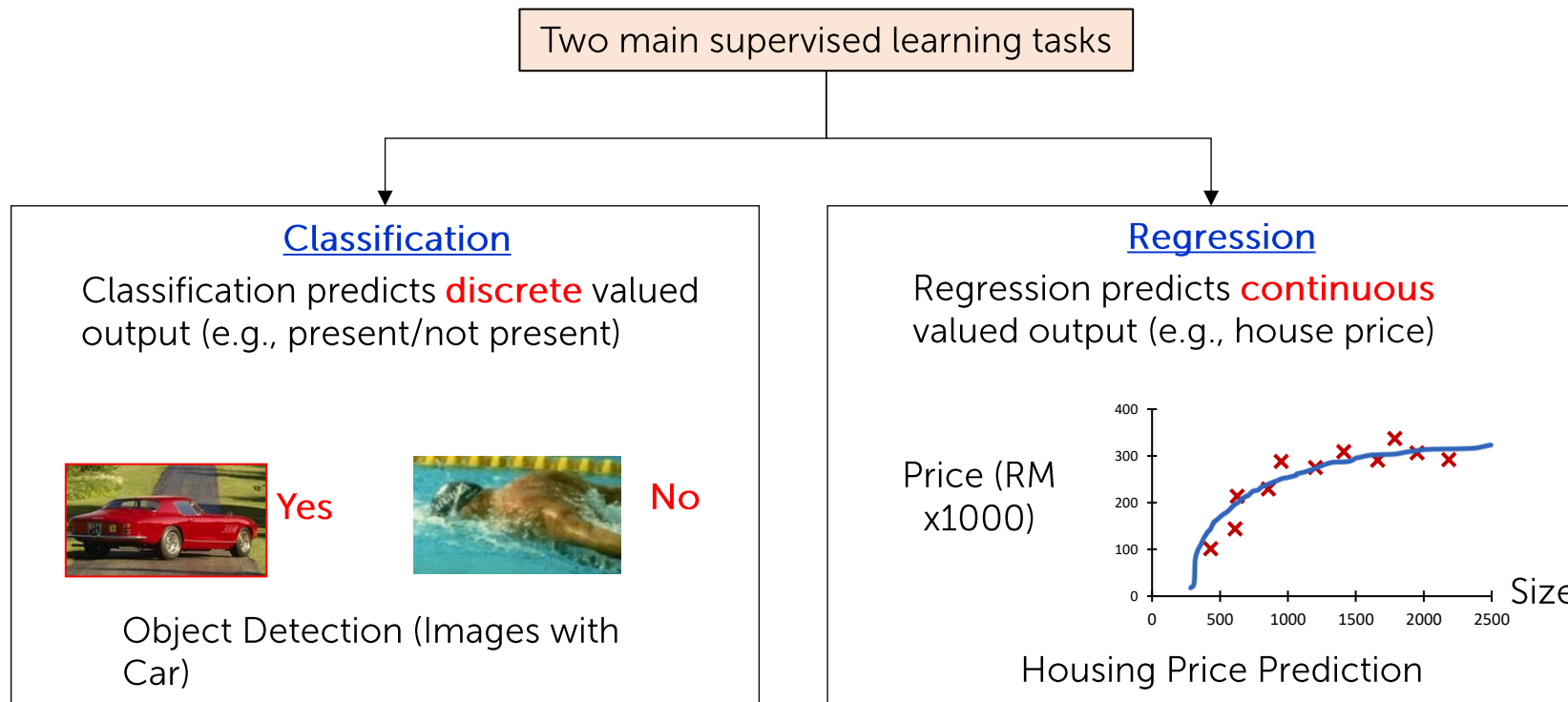


Source: <https://www.geeksforgeeks.org/supervised-machine-learning/>

Types of Supervised Learning

Key Points:

- Supervised learning involves training a machine from labeled data.
- Labeled data consists of examples with the correct answer or classification.
- The machine learns the relationship between inputs (fruit images) and outputs (fruit labels).
- The trained machine can then make predictions on new, unlabeled data.



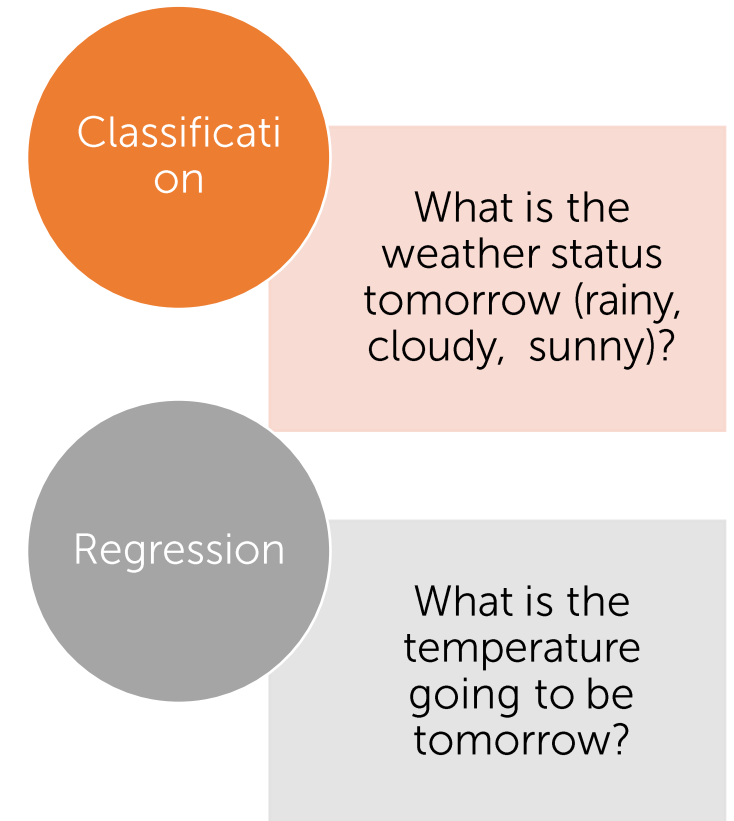
When to Use Regression/Classification?

Classification

- **Purpose:** Categorize datasets into discrete, mutually exclusive classes.
- **Typical Use:** Binary outcomes, such as "Yes" or "No."
- **Extensions:** Modified algorithms handle multi-class scenarios (e.g., predicting among "Red," "Blue," and "Green").
- **Example Applications:**
 - Determining whether a loan application is approved or not.
 - Predicting the type of fruit based on physical characteristics.

Regression

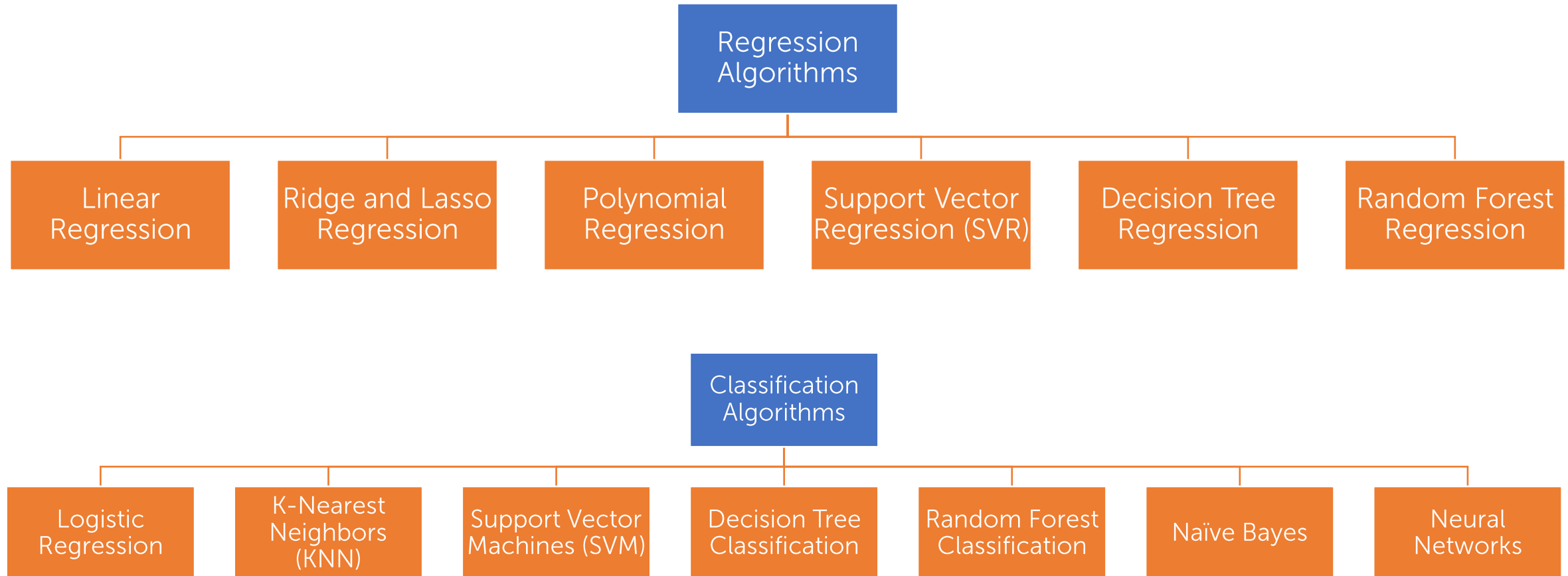
- **Purpose:** Predict continuous response variables.
- **Typical Use:** Continuous outcomes like numeric values.
- **Example Applications:**
 - Predicting house prices based on size, location, and features.
 - Estimating daily temperatures.



Classification vs Regression in Machine Learning

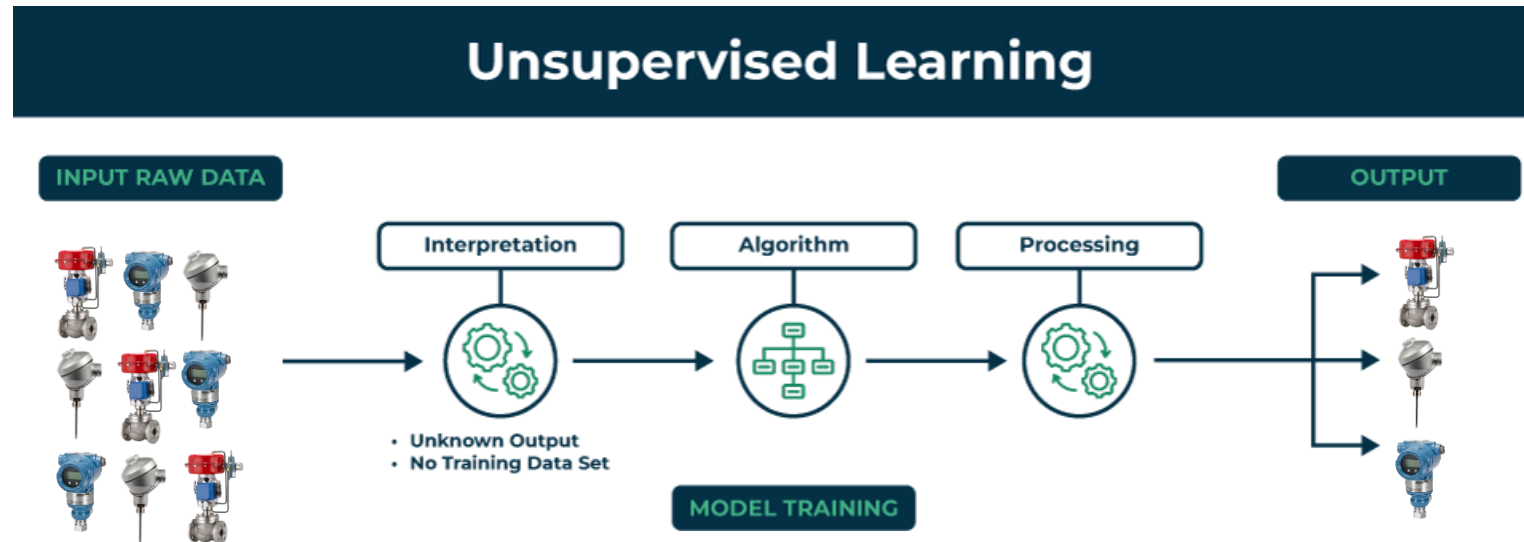
Classification	Regression
In this problem statement, the target variables are discrete.	In this problem statement, the target variables are continuous.
Problems like Spam Email Classification, Disease prediction like problems are solved using Classification Algorithms.	Problems like House Price Prediction, Rainfall Prediction like problems are solved using regression Algorithms.
Input Data are Independent variables and categorical dependent variable.	Input Data are Independent variables and continuous dependent variable.
The classification algorithm's task mapping the input value of x with the discrete output variable of y .	The regression algorithm's task is mapping input value (x) with continuous output variable (y).
Output is Categorical labels.	Output is Continuous numerical values.
Objective is to Predict categorical/class labels.	Objective is to Predicting continuous numerical values.

Supervised Learning Algorithms



Unsupervised Learning

- **Definition:** In unsupervised learning, the data is “unlabeled”, and the algorithm identifies patterns or structures within the data.
- **Goal:** To uncover hidden structures or groupings in the data.
- **Techniques:**
 - **Clustering:** Grouping similar data points (e.g., customer segmentation).
 - **Association rules:** Groups of items frequently occurring together, identified as a precursor to generating association rules.
- **Applications:**
 - Organizing news articles into topics.
 - Identifying fraudulent transactions in banking.

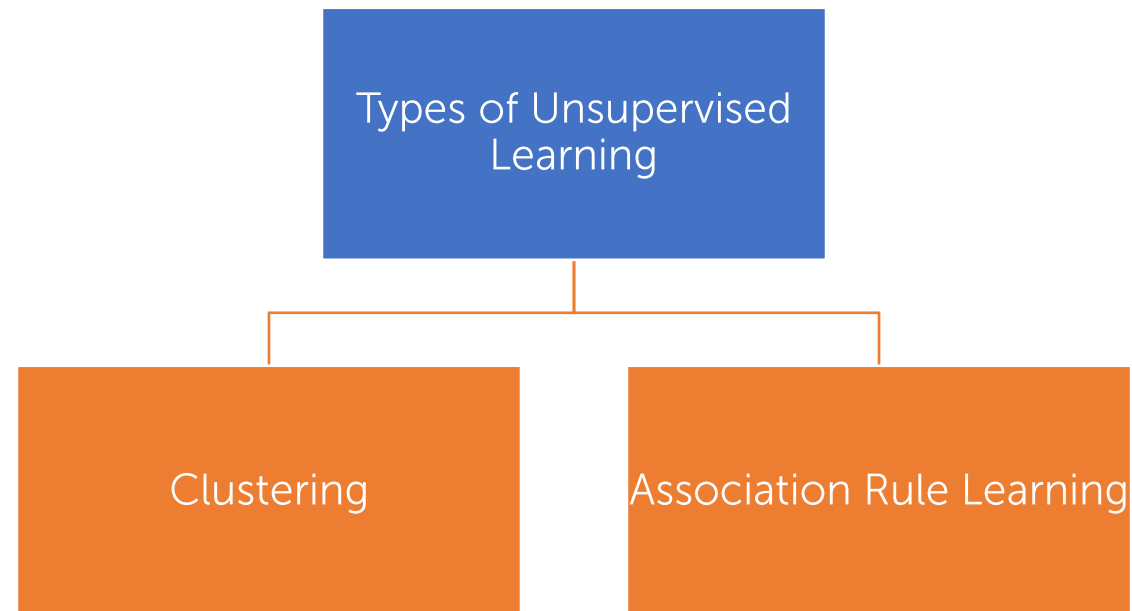


Source: <https://www.geeksforgeeks.org/unsupervised-learning/>

Types of Unsupervised Learning

Key Points:

- Unsupervised learning allows the model to discover patterns and relationships in unlabeled data.
- Clustering algorithms group similar data points together based on their inherent characteristics.
- Feature extraction captures essential information from the data, enabling the model to make meaningful distinctions.
- Label association assigns categories to the clusters based on the extracted patterns and characteristics.



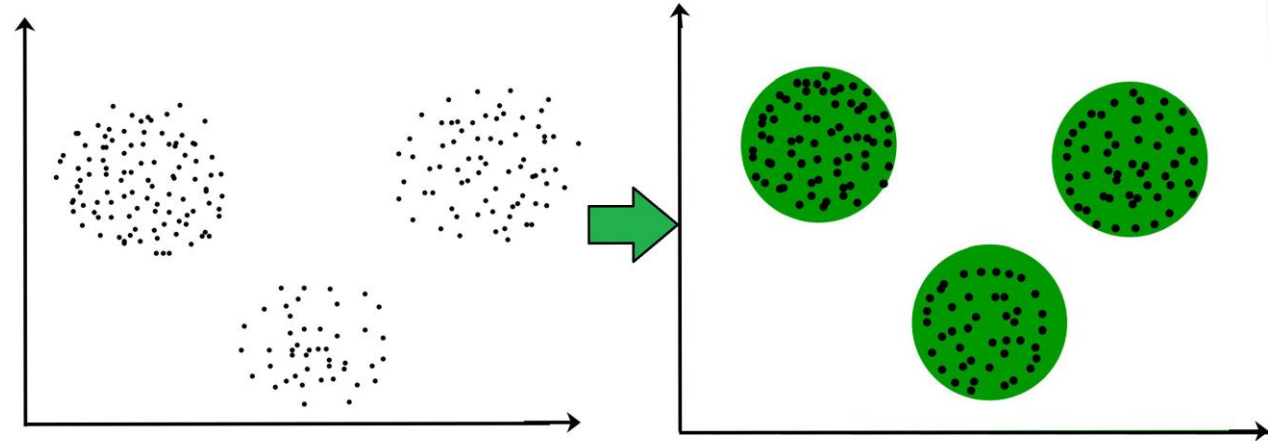
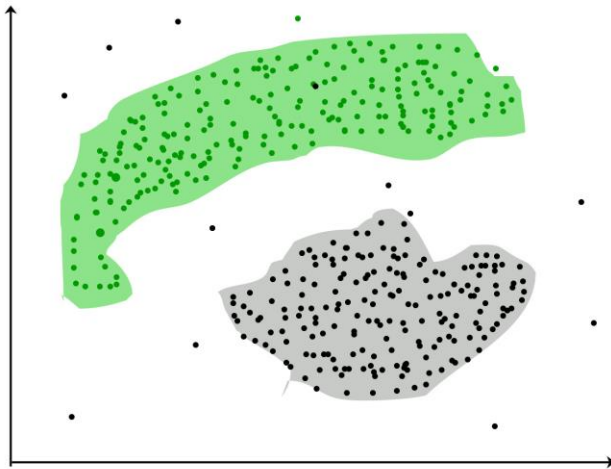
Clustering

The task of grouping data points based on their similarity with each other is called Clustering.

This method is defined under the branch of Unsupervised Learning, which aims to gain insights from unlabelled data points. Unlike supervised learning, we don't have a target variable.

Clustering aims at forming groups of homogeneous data points from a heterogeneous dataset.

It evaluates the similarity based on a metric like Euclidean distance, Cosine similarity, Manhattan distance, etc. and then group the points with highest similarity score together.



Source: <https://www.geeksforgeeks.org/clustering-in-machine-learning/>

Types of Clustering

There are 2 types of clustering that can be performed to group similar data points.

Hard Clustering:

In this type of clustering, each data point belongs to a cluster completely or not. For example, Let's say there are 4 data points, and we have to cluster them into 2 clusters.

So, each data point will either belong to cluster 1 or cluster 2.

Data Points	Clusters
A	C1
B	C2
C	C2
D	C1

Soft Clustering:

In this type of clustering, instead of assigning each data point into a separate cluster, the probability or likelihood of that point being that cluster is evaluated. For example, Let's say there are 4 data points, and we have to cluster them into 2 clusters.

So, we will evaluate the probability of a data point belonging to both clusters. This probability is calculated for all data points.

Data Points	Probability of C1	Probability of C2
A	0.91	0.09
B	0.3	0.7
C	0.17	0.83
D	1	0

Association Rule Learning

Association rule learning is a key technique in unsupervised learning used to uncover interesting relationships, patterns, or correlations among items in large datasets.

This method is particularly useful in analyzing transaction or market basket data to identify rules that explain how items or features are associated.

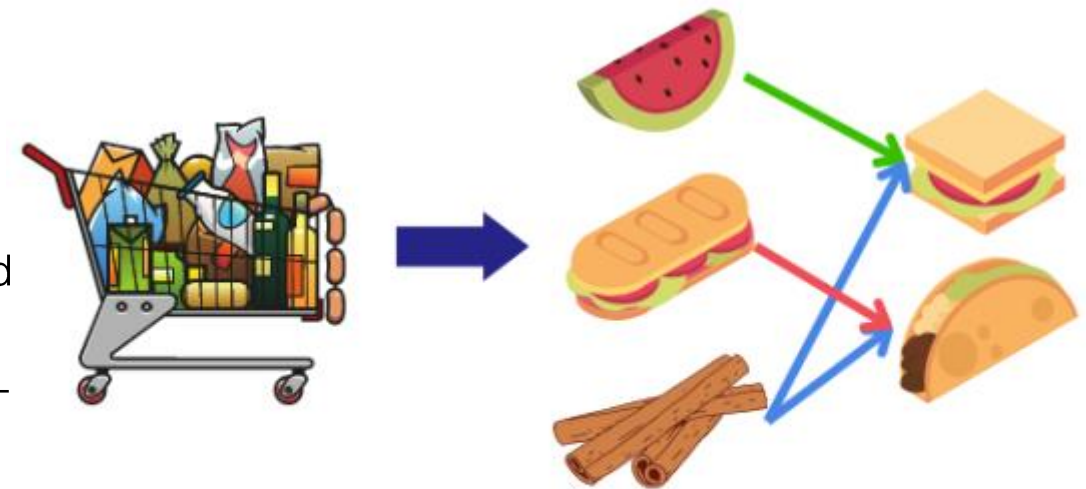
Key Concept

An association rule is an implication of the form:

$$A \Rightarrow B$$

- A (antecedent): The item(s) or condition(s) on the left-hand side of the rule.
- B (consequent): The item(s) or condition(s) on the right-hand side of the rule.

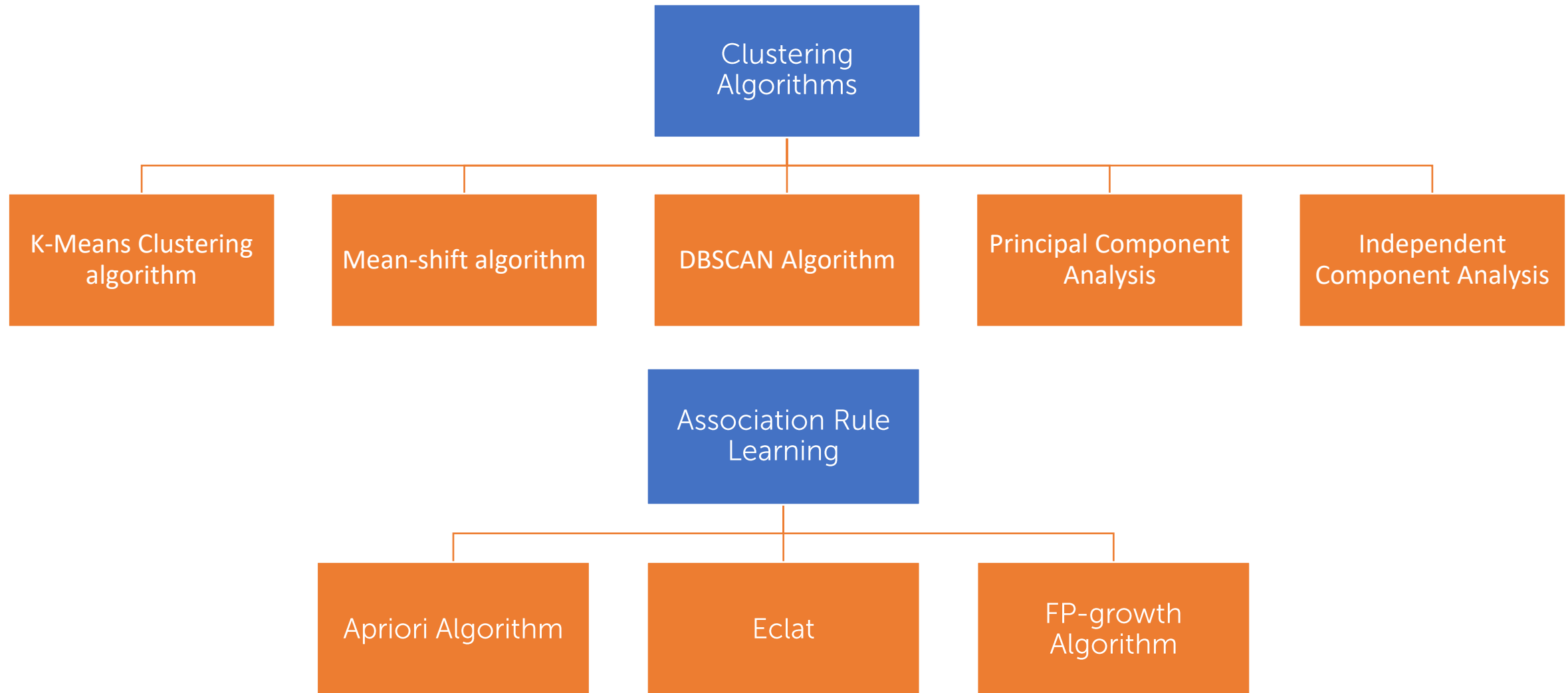
The rule implies that when A is present, B is likely to be present with a specific probability.



*"93% of people who purchased item A
also purchased item B"*

Source: <https://pythongeeks.org/unsupervised-machine-learning/>

Unsupervised Learning Algorithms



Semi-Supervised Learning

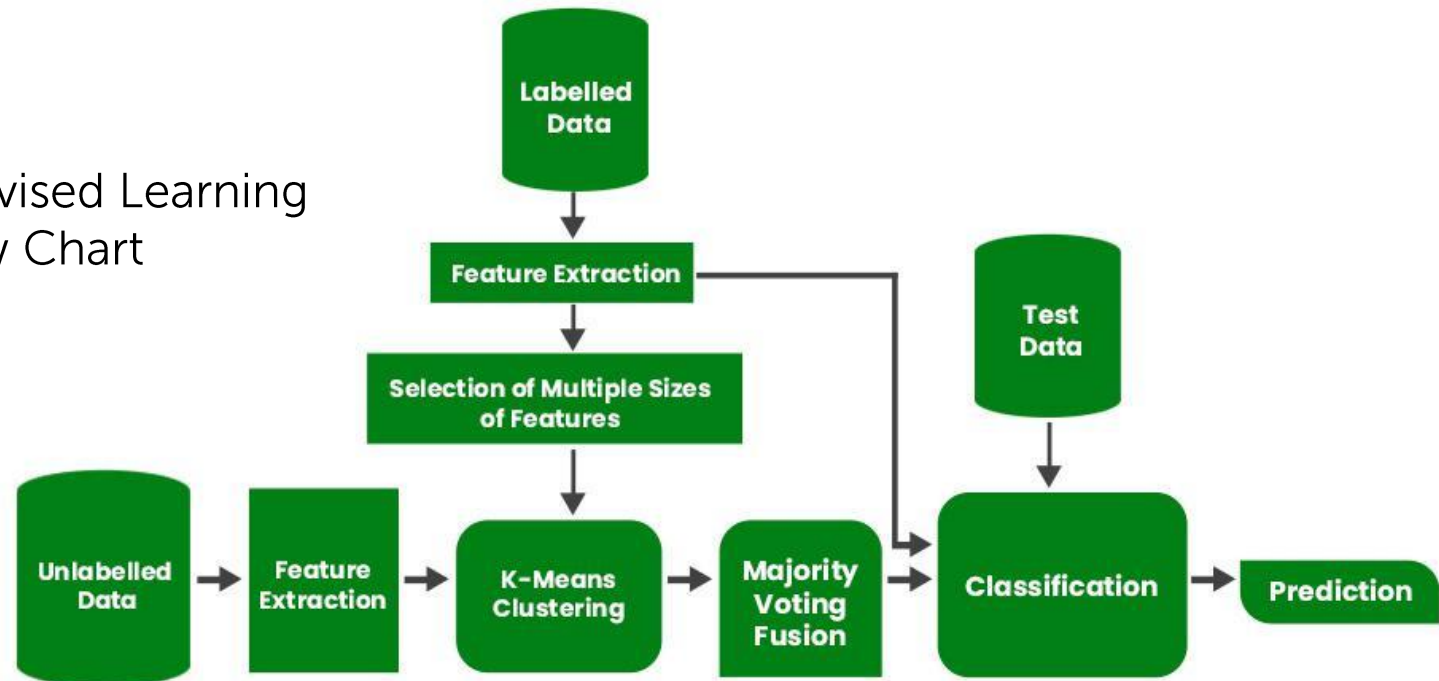
Semi-supervised learning is a machine learning algorithm that works between supervised and unsupervised learning, so it uses both labelled and unlabelled data.

It's particularly useful when obtaining labeled data is costly, time-consuming, or resource-intensive.

This approach is useful when the dataset is expensive and time-consuming.

Semi-supervised learning is chosen when labeled data requires skills and relevant resources in order to train or learn from it.

Semi-Supervised Learning
Flow Chart



Source: <https://www.geeksforgeeks.org/ml-semi-supervised-learning/>

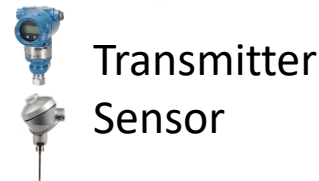
Semi-Supervised Learning

- **Definition:** The algorithm is trained on a dataset that is partially labelled and partially unlabeled. This approach is useful when labelling data is expensive or time-consuming.
- **Goal:** To leverage a small amount of labelled data and a large amount of unlabeled data to improve learning.
- **Techniques:**
 - Combines methods from supervised and unsupervised learning.
- **Applications:**
 - Image recognition with limited labelled examples.
 - Speech analysis when only some data is transcribed.
 - Webpage classification.

Input Data

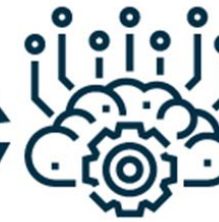


Partial Labels



Transmitter
Sensor

Machine Learning Model



Prediction

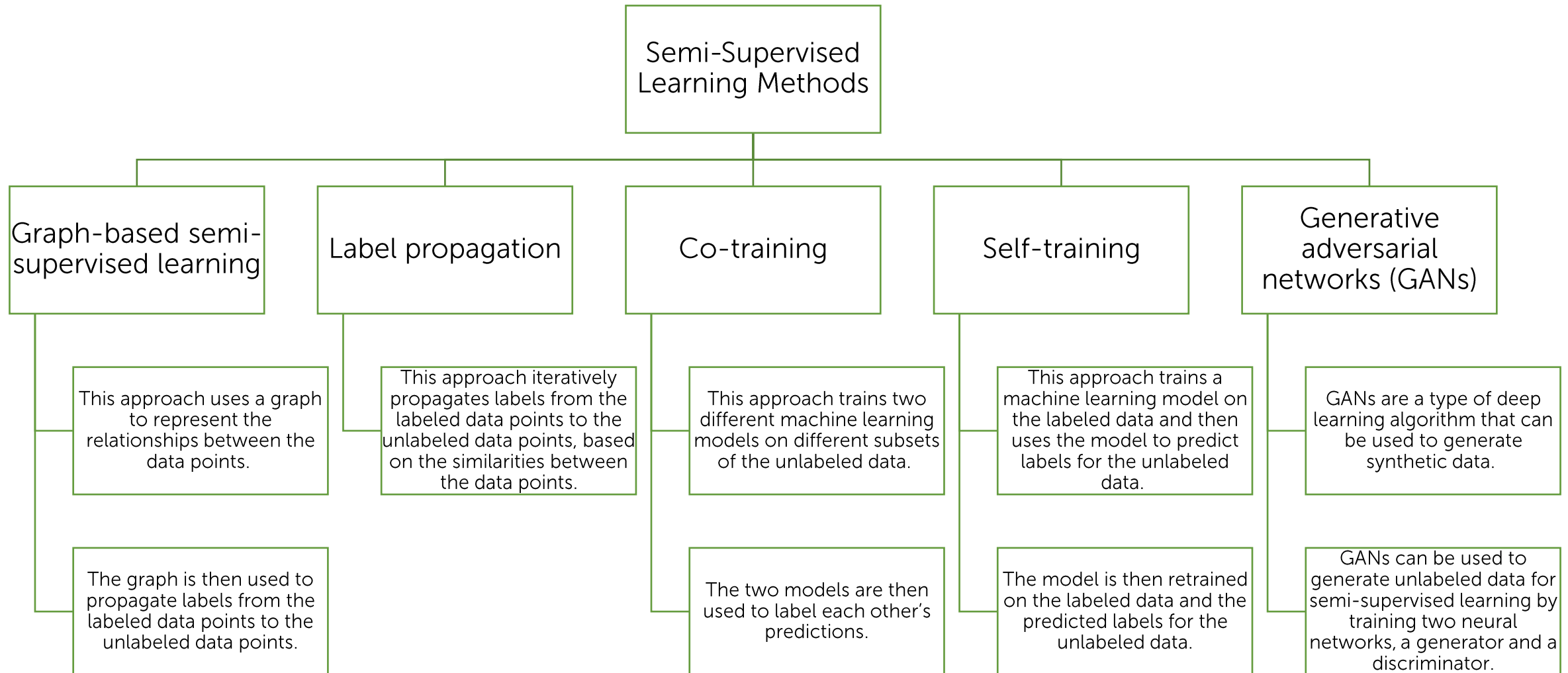
It's a Control Valve

Unlabelled Data



Source: <https://www.geeksforgeeks.org/ml-semi-supervised-learning/>

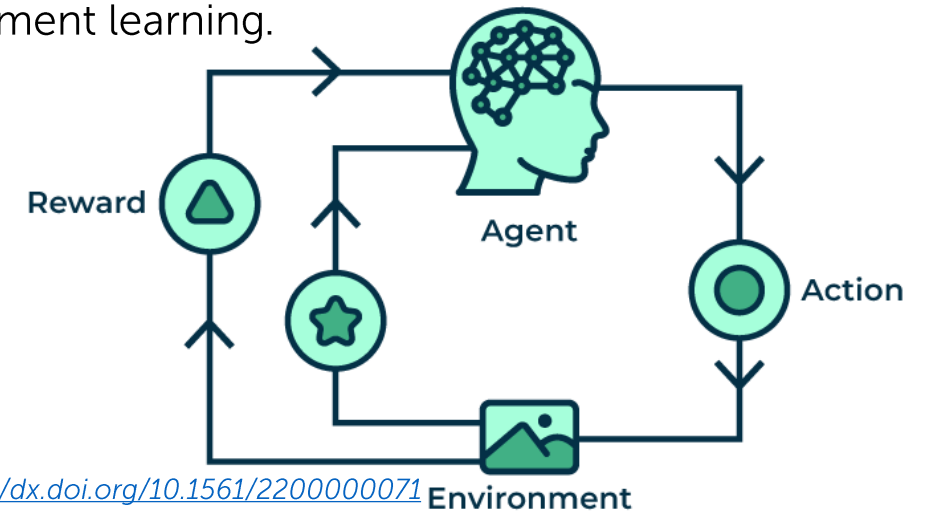
Types of Semi-Supervised Learning Methods



Reinforcement Machine Learning

Reinforcement machine learning algorithm is a learning method that interacts with the environment by producing actions and discovering errors.

Trial, error, and delay are the most relevant characteristics of reinforcement learning.



In this technique, the model continuously increases its performance using Reward Feedback to learn the behavior or pattern.

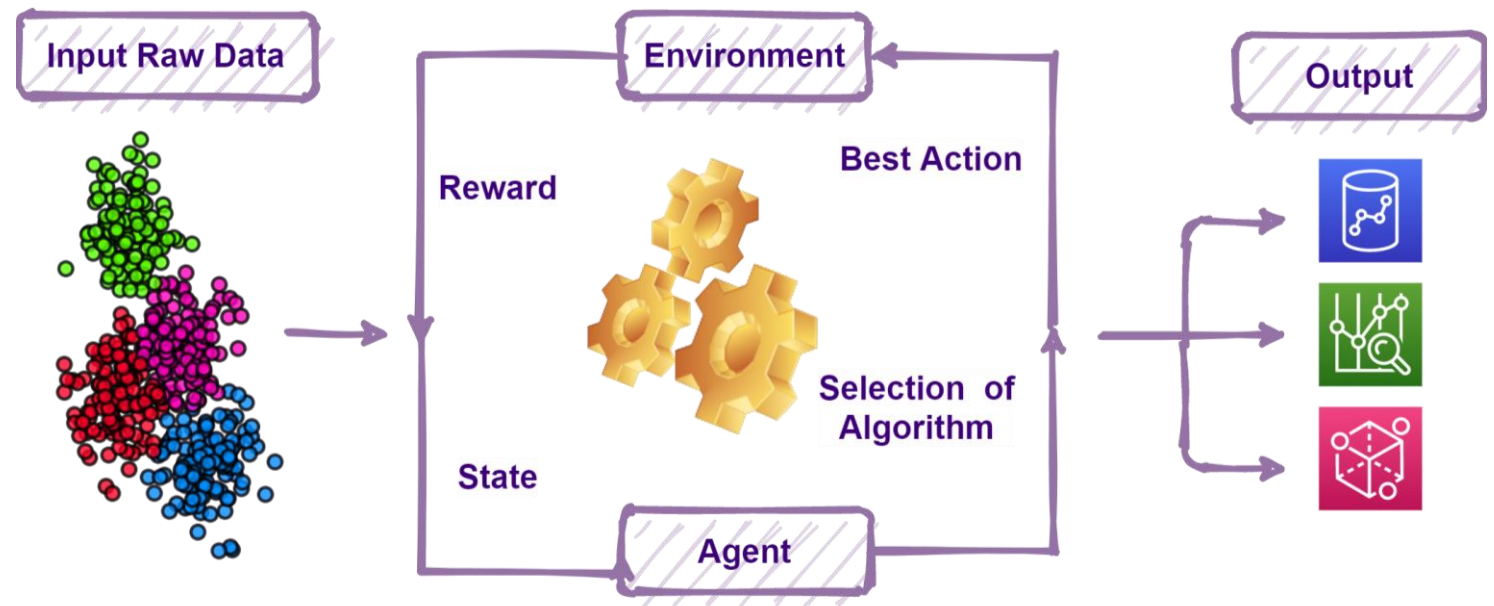
These algorithms are specific to a particular problem, e.g. Google Self Driving car, AlphaGo, where a bot competes with humans and even itself to get better and better performers in Go Game.

Each time we feed in data, they learn and add the data to their knowledge which is training data.

So, the more it learns the better it gets trained and hence experienced.

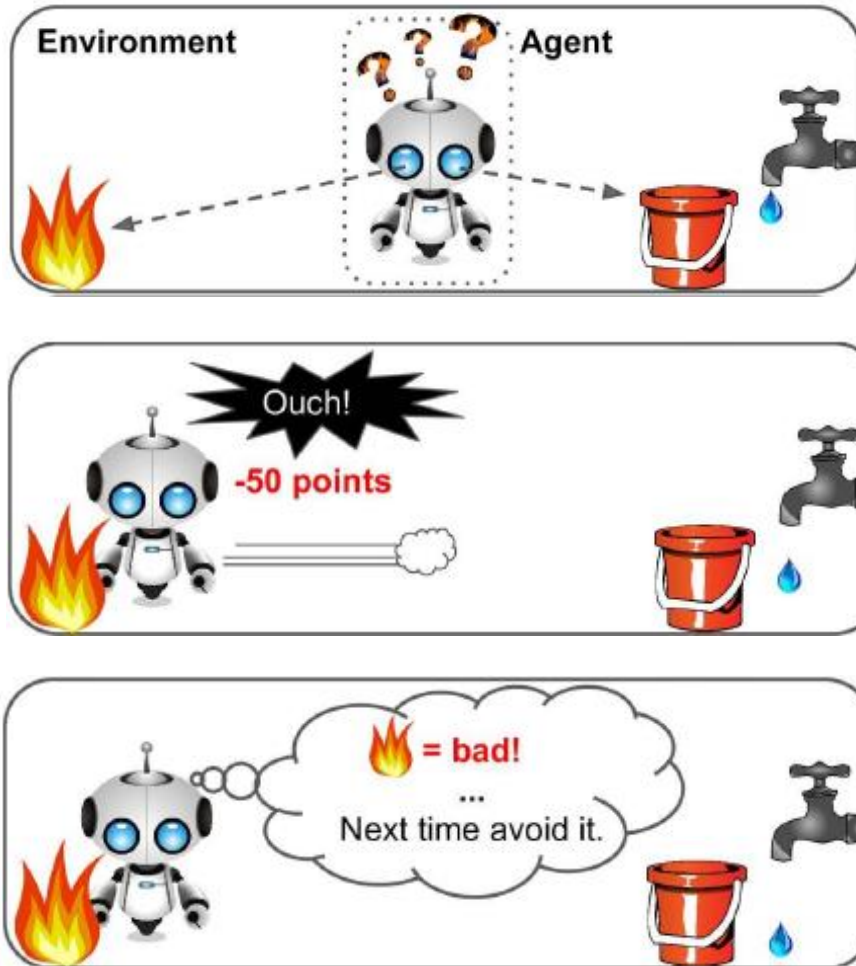
Reinforcement Machine Learning

- **Definition:** The algorithm learns by interacting with an environment and receiving feedback through rewards or penalties.
- **Goal:** To find the optimal action sequence that maximises cumulative rewards.
- **Techniques:**
 - Q-learning.
 - Deep Reinforcement Learning.
- **Applications:**
 - Game playing (e.g., AlphaGo).
 - Robotics (e.g., robotic arms learning tasks).
 - Autonomous vehicles (e.g., optimizing navigation).



Source: <https://techvidvan.com/tutorials/reinforcement-learning/>

Reinforcement Machine Learning

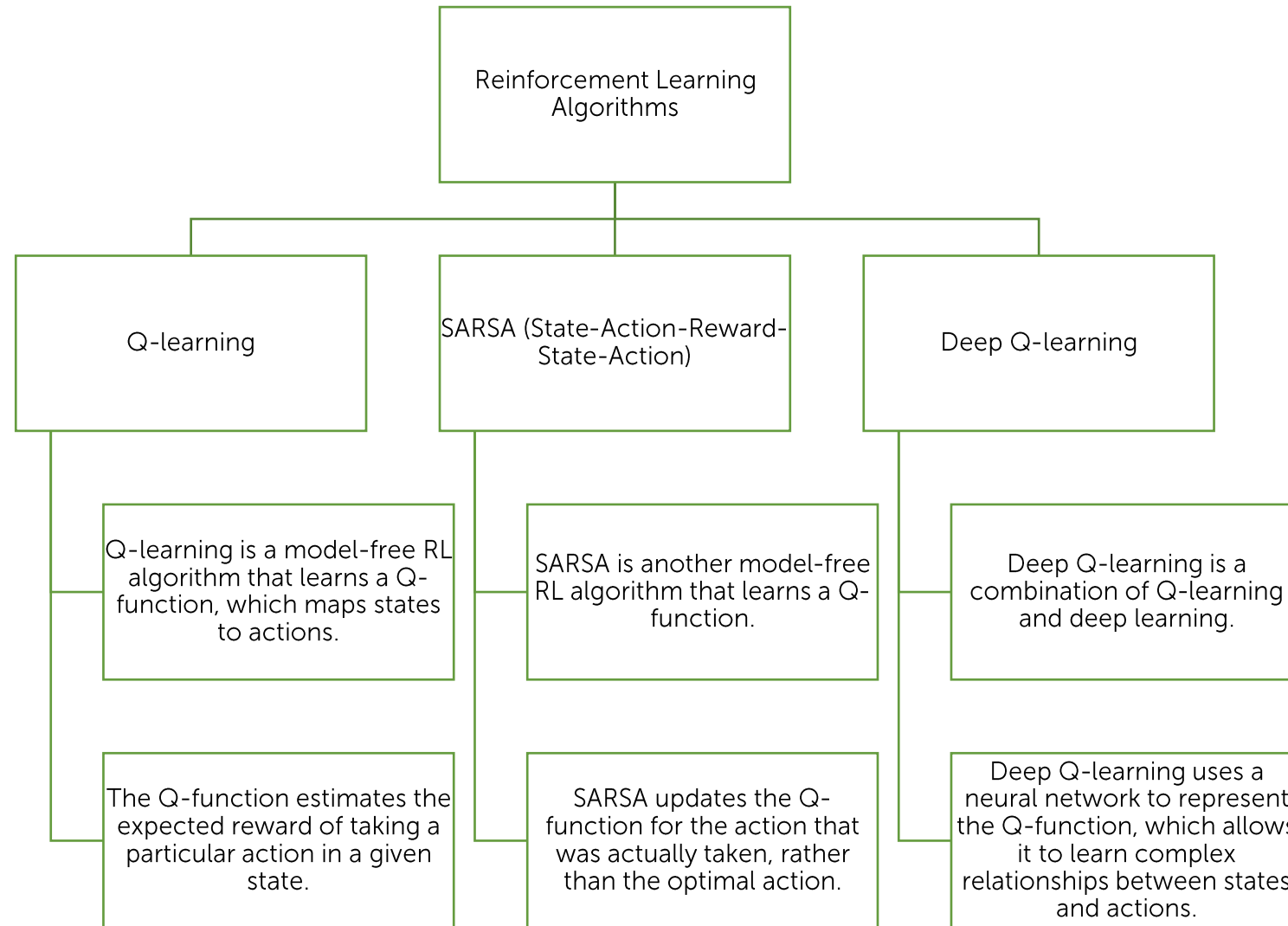


The agent

1. observes the environment
2. Select action using policy
3. Perform action
4. Get reward or penalty
5. Update policy (learning step)
6. Iterate until an optimal policy is found

Source: <http://dx.doi.org/10.1561/22000000071>

Types of Reinforcement Learning Algorithms



18.1 Introduction to ML

- Features of Machine Learning
- Types of Machine Learning
- **Differences between Machine Learning Types**
- Machine Learning Pipeline
- Machine Learning Framework
- Advantages and Disadvantages of ML

Differences between Machine Learning Types

Type	Input Data	Goal	Key Techniques	Example Applications
Supervised Learning	Labelled	Predict outcomes	Regression, Classification	Email spam detection, Forecasting
Unsupervised Learning	Unlabelled	Discover patterns or groups	Clustering, Association Rule Learning	Customer segmentation, Clustering
Semi-Supervised Learning	Partially labelled	Leverage unlabeled data for learning	Graph-based semi-supervised learning, Generative adversarial networks (GANs)	Image recognition, Text analysis
Reinforcement Learning	Interaction with environment	Maximize cumulative rewards	Q-learning, Deep Reinforcement Learning	Game AI, Autonomous vehicles

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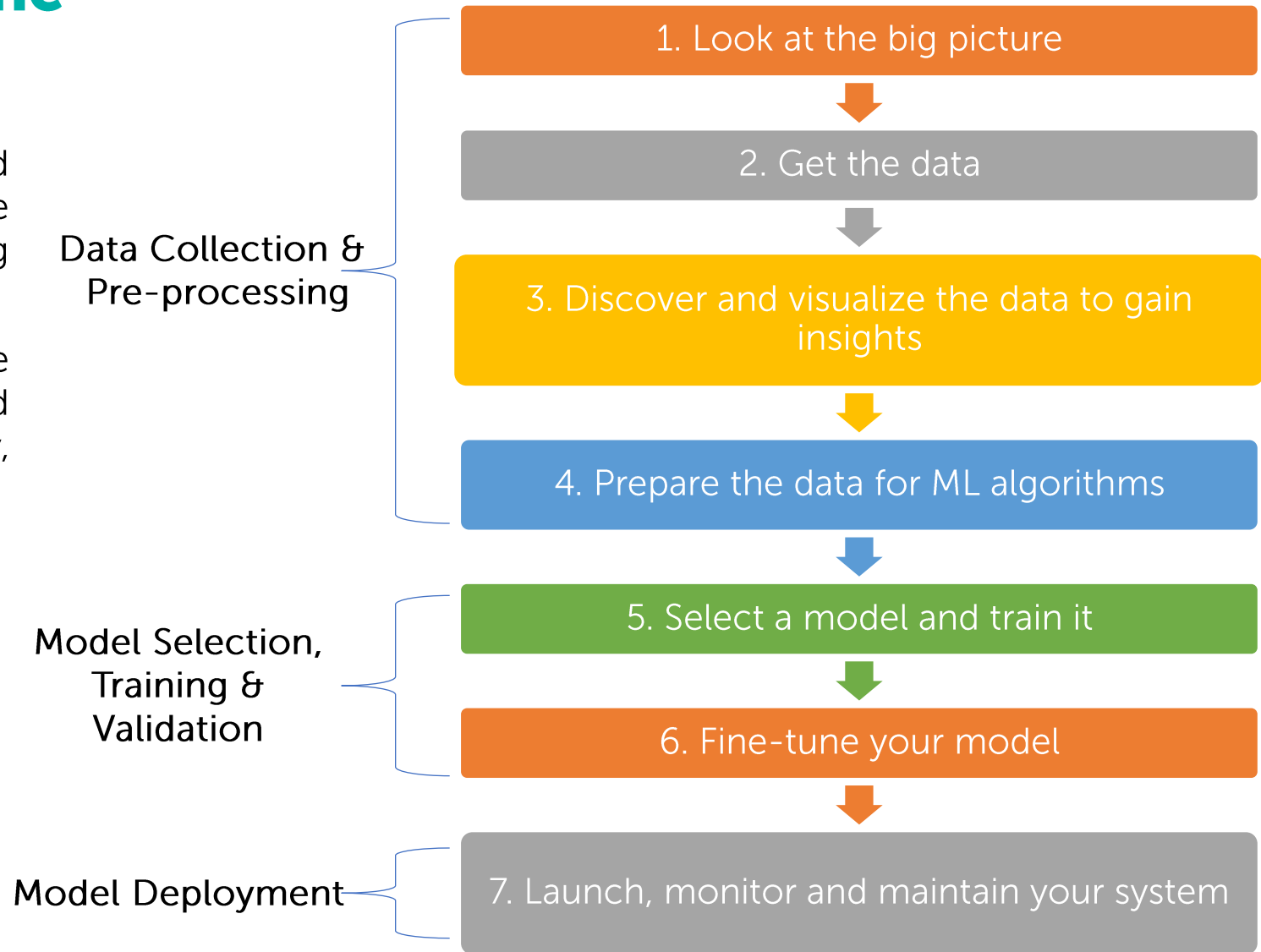
Machine Learning Pipeline

A machine learning pipeline is a structured workflow that automates and streamlines the process of building, training, and deploying machine learning models.

It ensures that each stage of the machine learning process is well-defined and seamlessly integrated, allowing for scalability, reproducibility, and efficiency.

Key Stages of a Machine Learning Pipeline

- Data Collection & Pre-processing
- Model Selection, Training & Validation
- Model Deployment



18.1 Introduction to ML

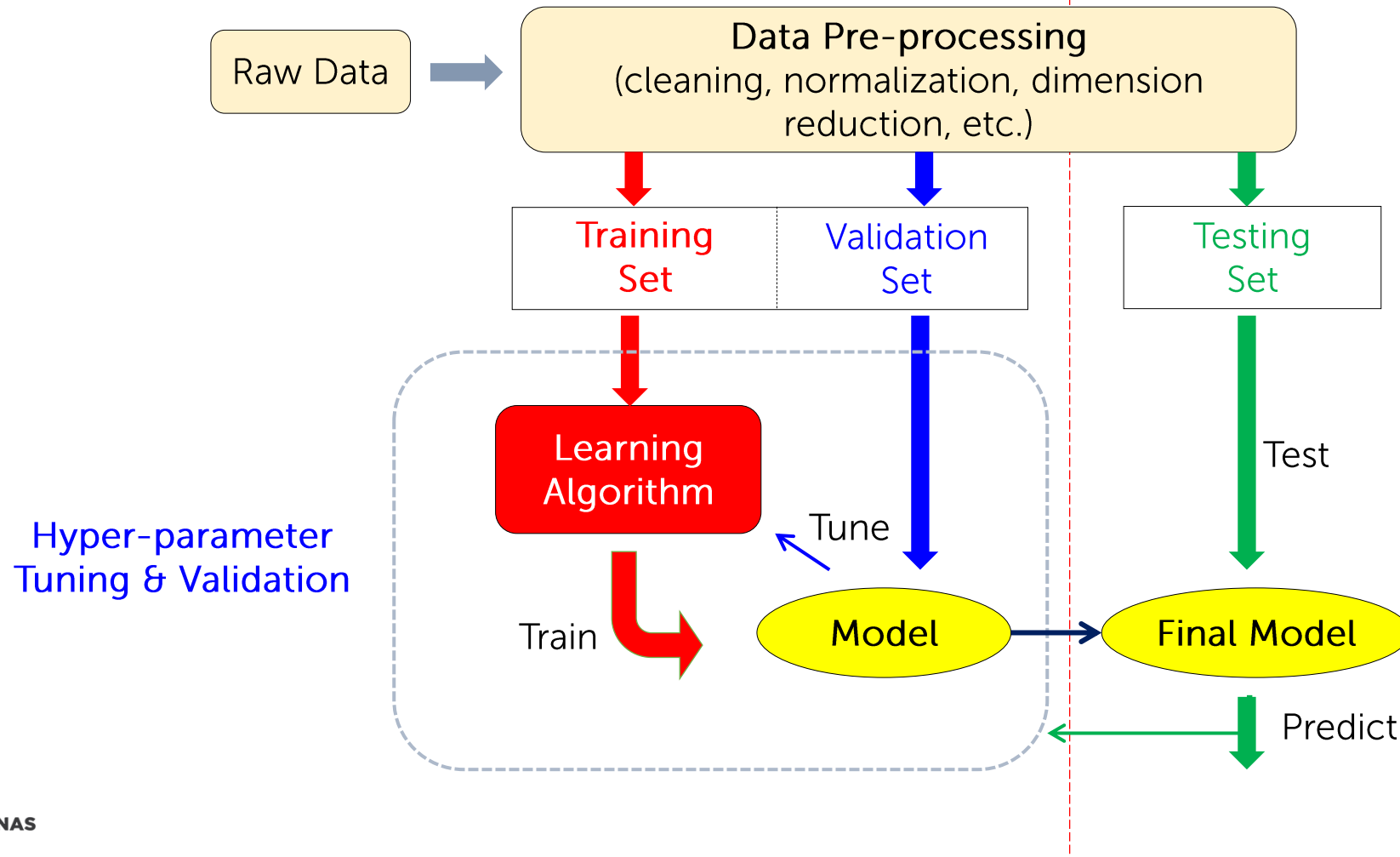
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Machine Learning Framework

Divided into two phases

Training Phase:

Testing Phase:



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- **Advantages and Disadvantages of ML**

Advantages and Disadvantages of ML

Advantages	Disadvantages
Identifies trends and patterns: ML analyses large datasets, like Amazon's product recommendations, to spot trends.	Data Acquisition: ML requires large, high-quality, and unbiased data, which can be hard to obtain.
Automation: ML models make predictions and improve autonomously, reducing human intervention (e.g., anti-virus software).	Time and Resources: Training ML models demands significant time and computational power.
Continuous Improvement: ML models improve in accuracy with more data, enhancing decision-making (e.g., weather forecasting).	Interpretation of Results: Interpreting ML results can be complex and requires choosing the right algorithms.
Handles complex data: ML processes multi-dimensional, diverse data in dynamic environments.	High error-susceptibility: ML models are sensitive to errors, especially with biased or incomplete data.
Wide Applications: ML benefits industries like e-commerce, healthcare, and academia, offering personalized experiences.	Need for Supervision: ML models need regular updates and re-training, which can be time-consuming and costly.

18.2 Introduction to DL

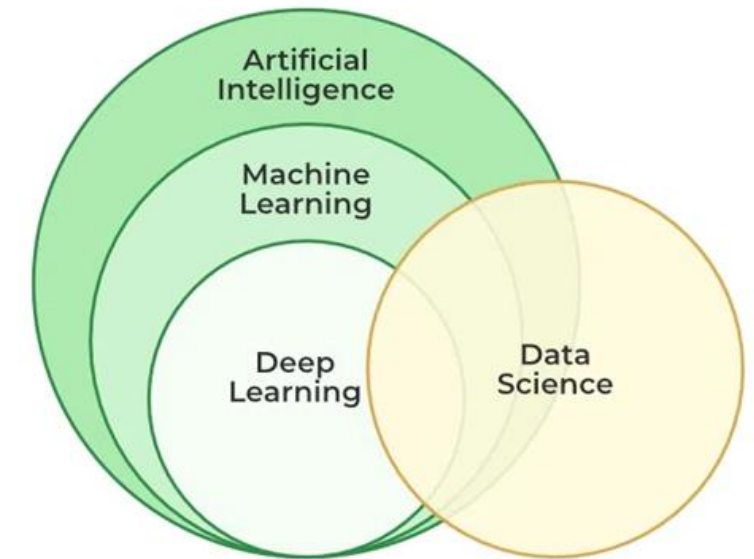
- Artificial Neural Networks (ANNs)
- How ANNs Work?
- Types of Activation Functions
- How does Deep Learning work?
- Types of Neural Networks
- Advantages and Disadvantages of DL

Introduction to DL

Deep Learning is a subset of Machine Learning (ML) that uses **Artificial Neural Networks (ANN)** with multiple layers to model and understand complex patterns in large datasets.

It mimics the way humans process information, enabling computers to perform tasks such as

- **Computer Vision:**
 - Image recognition, object detection, and facial recognition.
 - Example: Autonomous vehicles.
- **Natural Language Processing (NLP):**
 - Sentiment analysis, machine translation, and chatbots.
 - Example: Google Translate.
- **Speech Recognition:**
 - Voice assistants like Siri and Alexa use deep learning for speech-to-text conversion.



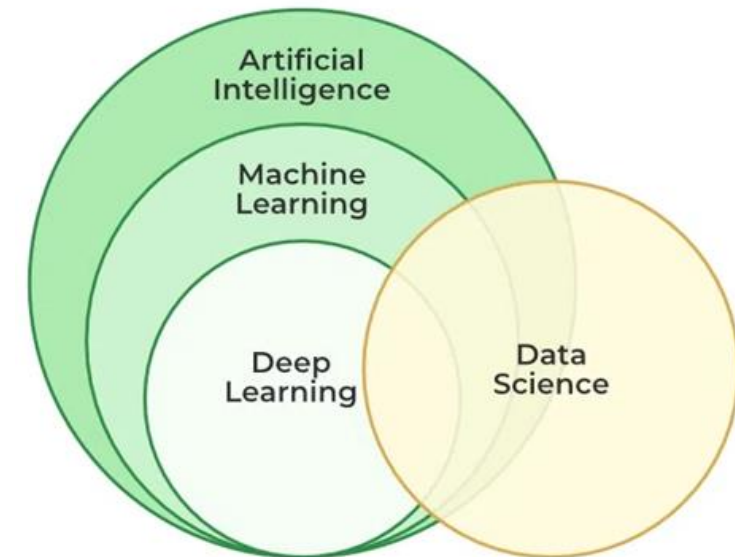
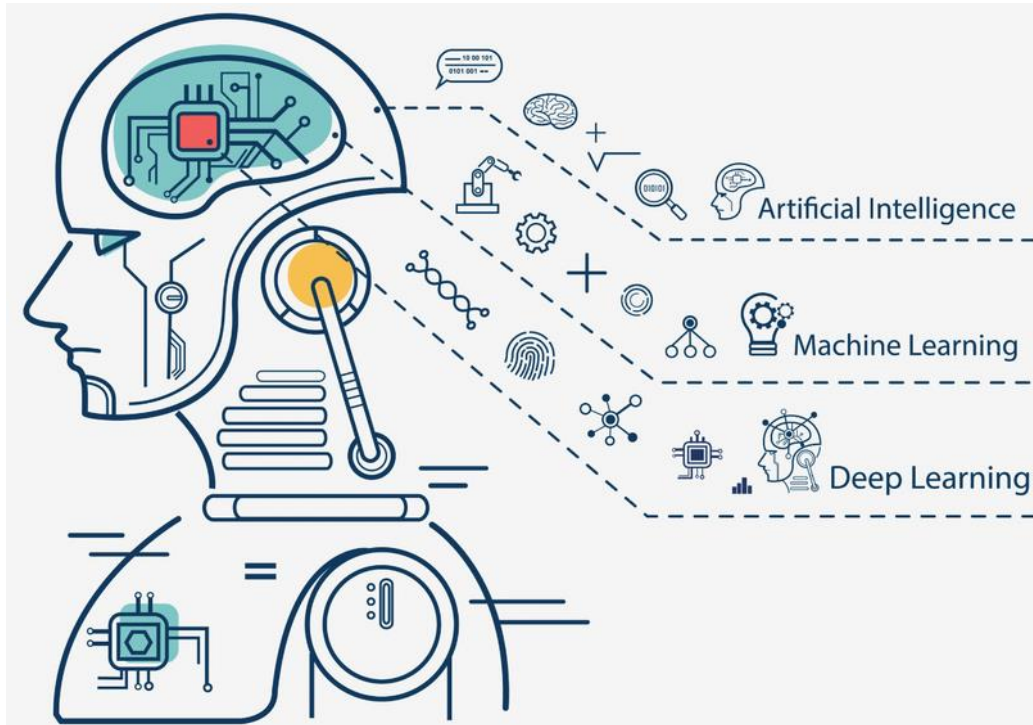
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Artificial Neural Networks (ANNs)

Deep Learning models use Artificial Neural Networks (ANNs) to extract information.

It is sometimes also referred to as a Deep Neural Network, where the term 'Deep' relates to the number of hidden layers in the neural network.

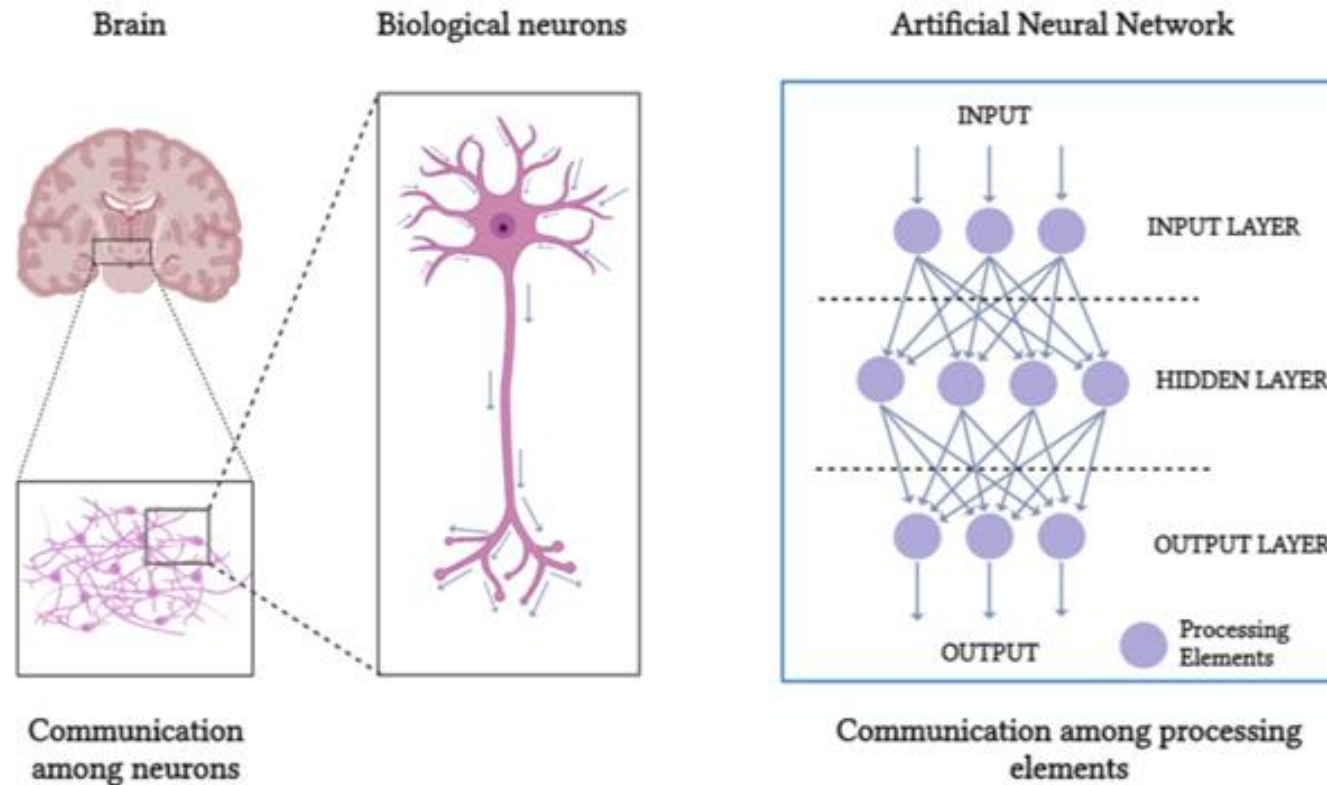


Artificial Neural Networks (ANNs)

An Artificial Neural Network is a network of biological neurons.

Deep learning is the process of teaching computers to process data in the same way that neural networks in the human brain do.

To mimic our brains, Artificial Neural Networks contain artificial neurons (or nodes).



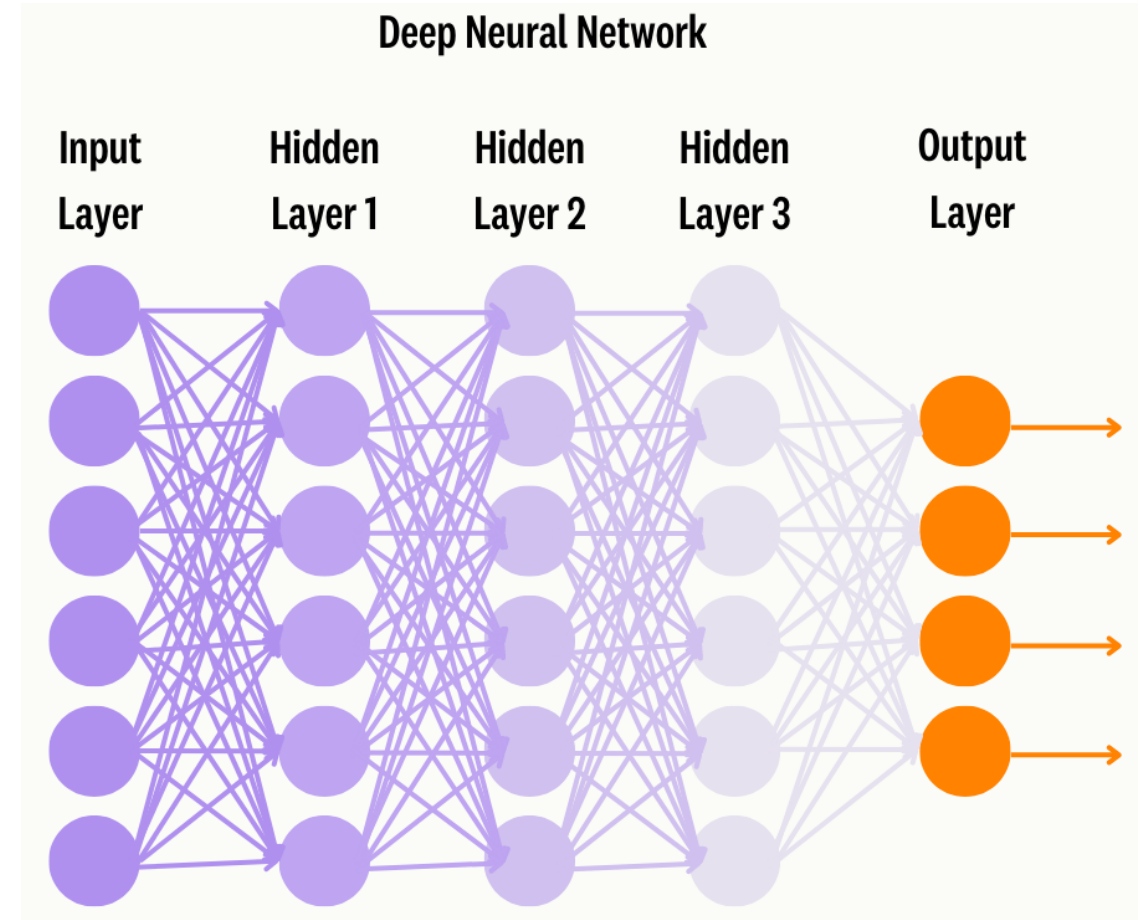
Artificial Neural Networks (ANNs)

Artificial Neural Networks (ANNs) comprise neurons containing three layers: an input layer, one or more hidden layers, and an output layer.

Each of these neurons is connected to another neuron, where computation happens.

- **Input Layer** - receives the input data.
- **Hidden Layer(s)** - perform mathematical computations on the input data
- **Output Layer** - returns the output data.

An artificial Neural network must have more than one hidden layer to be considered a deep learning network.



Source: <https://doi.org/10.1201/9781315273570>

18.2 Introduction to DL

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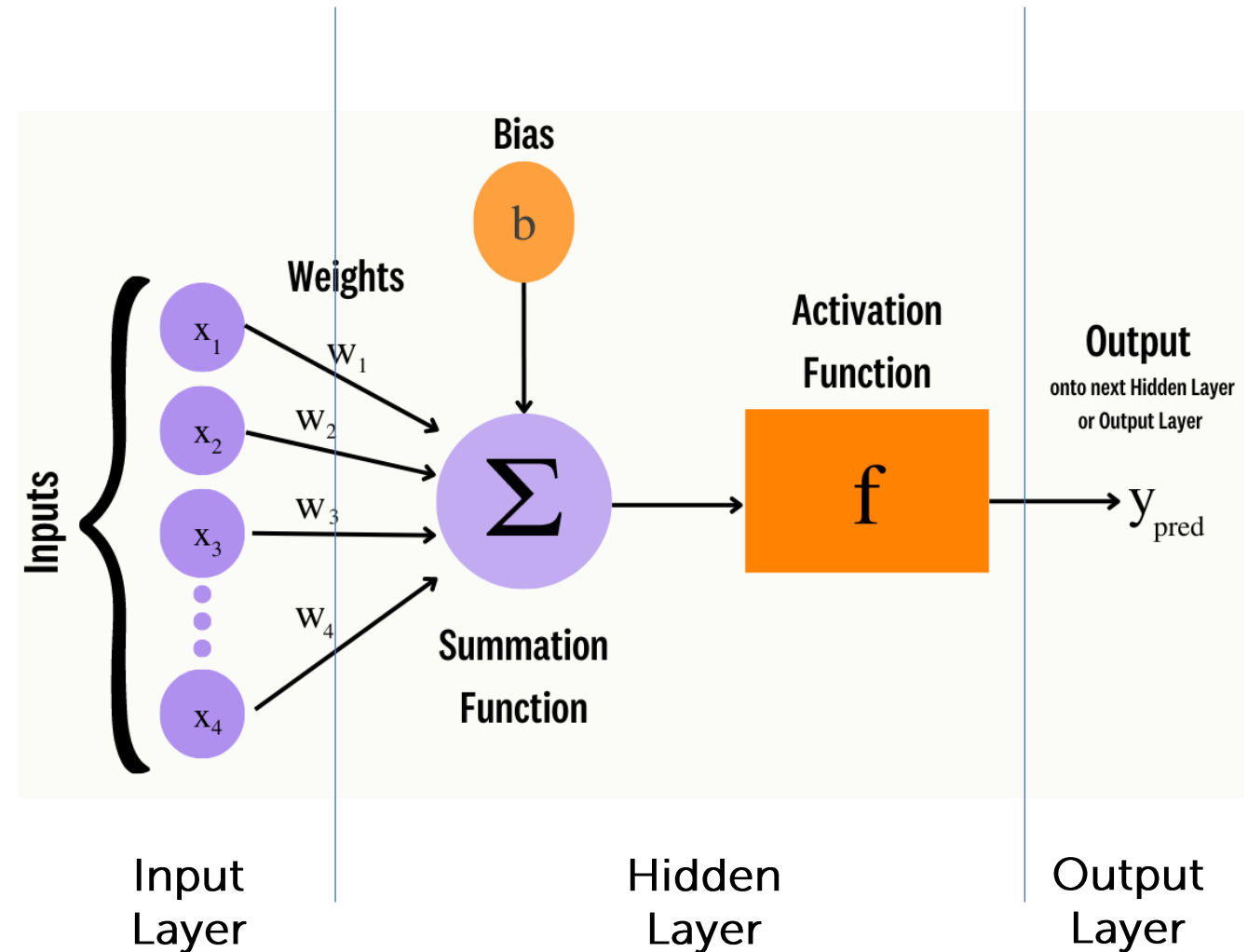
How ANNs Work?

Weight and Bias

- **Weight** controls the strength of the connection between two neurons. It decides how much influence the input has on the output.
- **Bias** will always have a value of 1 and is an additional input into the next layer. Bias aims to guarantee that there will always be activation in the neurons, even if the input is 0.

Hidden Layer

- The Hidden layer(s) brings your Deep Neural Network to life. These layers receive input from either the input layer or another hidden layer. Hidden layers also provide the output to other hidden or output layers.
- The reason these layers are called “hidden” is that we don’t know the true values of their nodes in the training dataset.

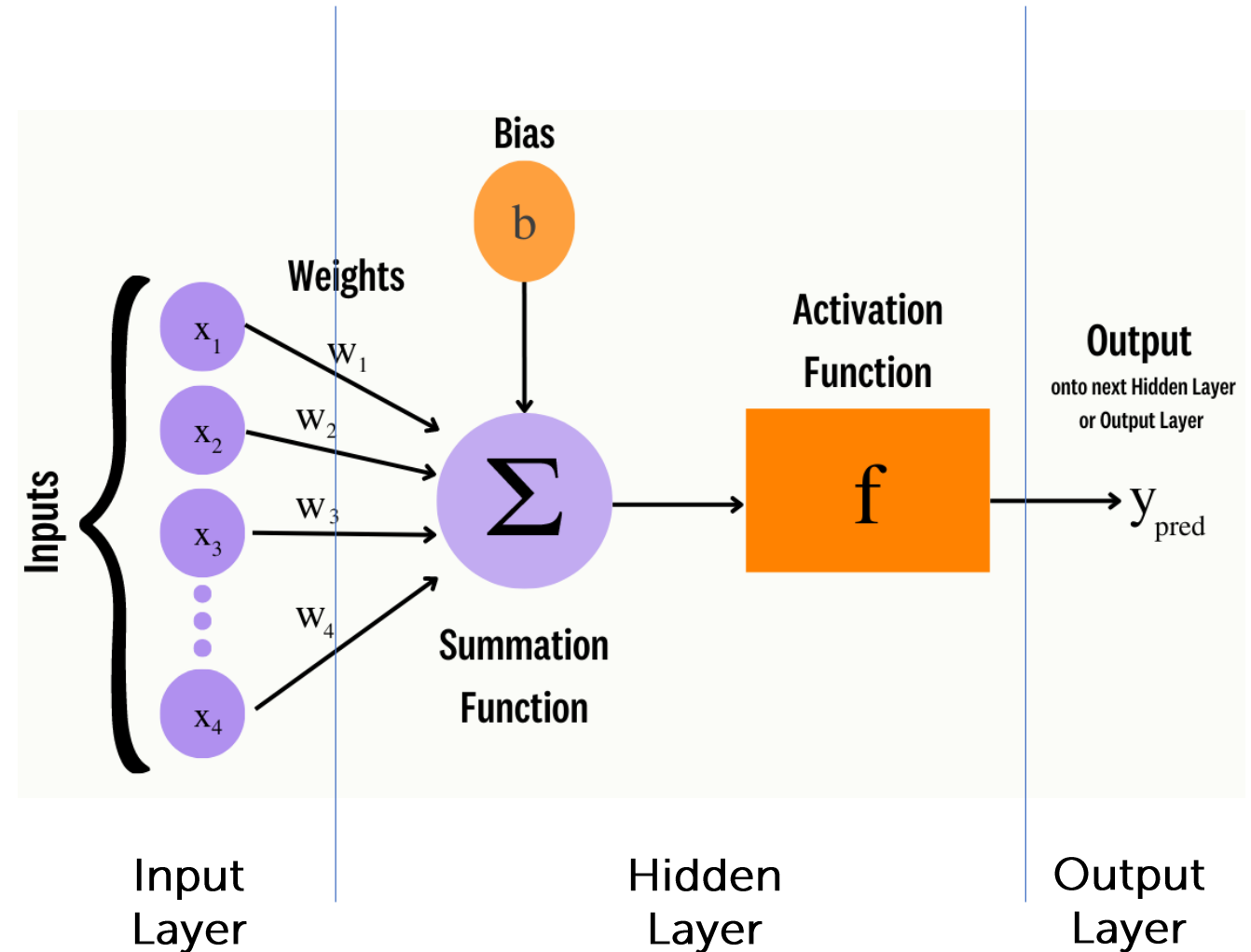


Source: <https://doi.org/10.1201/9781315273570>

How ANNs Work?

Activation Function

- An Activation Function decides whether a neuron should be activated or not.
- It uses mathematical concepts to decide whether a neuron's input will make a difference to the network or not.
- It can also be called a 'Transfer Function' as it transfers the next move.
- Activation functions are very beneficial to neural networks as they help them learn complex patterns in data and decide what pattern and information are vital to be fired to the next neuron.
- They control how well the neural network model learns on the training dataset.



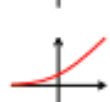
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Types of Activation Functions

- It is important to note that there are different types of Activation Functions, so choosing an Activation Function greatly impacts the neural network's performance.
- You can use different Activation Functions for different parts of the model to help achieve better performance levels.

Activation	Definition	Derivative	Range	Smoothness
 Identity (Id)	$g(x) = x$	$g'(x) = 1$	$\mathbb{R}$	C^∞
 Sigmoid (Logistic, σ)	$g(x) = \frac{1}{1 + e^{-x}}$	$g'(x) = g(x)(1 - g(x))$	$(0, 1)$	C^∞
 Heaviside (H)	$g(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$	$g'(x) = 0, x \neq 0$	$[0, 1]$	none
 Hyperbolic tangent (tanh)	$g(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$	$g'(x) = 1 - g(x)^2$	$(-1, 1)$	C^∞
 Rectified linear unit (ReLU)	$g(x) = \max\{x, 0\}$	$g'(x) = \begin{cases} 0, & x < 0 \\ 1, & x > 0 \end{cases}$	$[0, \infty)$	C
 Parametric rectified linear unit (PReLU)	$g(x) = \begin{cases} \alpha x, & x < 0 \\ x, & x \geq 0 \end{cases}$ $\alpha > 0$	$g'(x) = \begin{cases} \alpha, & x < 0 \\ 1, & x > 0 \end{cases}$	$\mathbb{R}$	C
 Exponential linear unit (ELU)	$g(x) = \begin{cases} \alpha(e^x - 1), & x < 0 \\ x, & x \geq 0 \end{cases}$ $\alpha > 0$	$g'(x) = \begin{cases} g(x) + \alpha, & x < 0 \\ 1, & x > 0 \end{cases}$	$(-\alpha, \infty)$	$C^1, \alpha = 1$ $C, \alpha \neq 1$
 Softplus	$g(x) = \log(1 + e^x)$	$g'(x) = \frac{1}{1 + e^{-x}}$	$(0, \infty)$	C^∞
 Gaussian	$g(x) = e^{-x^2}$	$g'(x) = -2xg(x)$	$(0, 1]$	C^∞

How to choose the right Activation Function?

- Activation Functions are used to introduce non-linearity in the network.
- A neural network will almost always have the same activation function in all hidden layers. This activation function should be differentiable so that the parameters of the network are learned in backpropagation.
- ReLU is the most commonly used activation function for hidden layers.

A few rules for choosing the activation function for your output layer based on the type of prediction problem that you are solving:

- Regression - Linear Activation Function
- Classification—Sigmoid or Softmax

The activation function used in hidden layers is typically chosen based on the type of neural network architecture.

- Convolutional Neural Network (CNN): ReLU activation function.
- Recurrent Neural Network: Tanh and/or Sigmoid activation function.

18.2 Introduction to DL

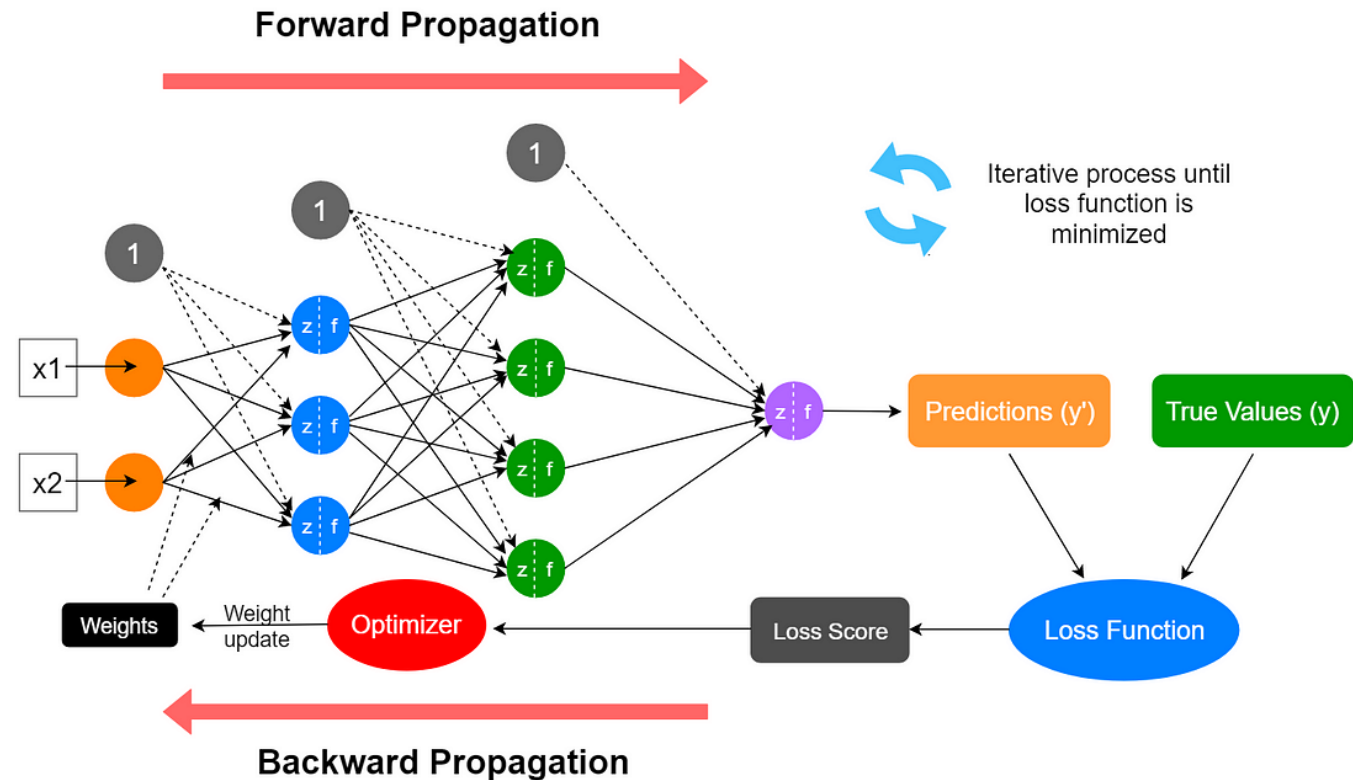
- Artificial Neural Networks (ANNs)
- How ANNs Work?
- Types of Activation Functions
- **How does Deep Learning work?**
- Types of Neural Networks
- Advantages and Disadvantages of DL

How does Deep Learning work?

Now, let's dive into how Deep Learning works.

1. The input layer is the first layer to receive data. Weights are assigned to each of the data variables
2. These variables are multiplied by their respective weights and then summed up.
3. Bias will then be added to the summed value.
4. The summed value is then passed through an activation function, which will determine the output.
5. If the output meets or exceeds the specific threshold, the neuron is activated, and the data is passed from one connected layer to another in the network.

If the neuron is activated, the output becomes the input for the next neuron, or if it is at the end of the process, it is the overall output.



Source: <https://doi.org/10.1007/978-3-319-73004-2>

How does Deep Learning work?

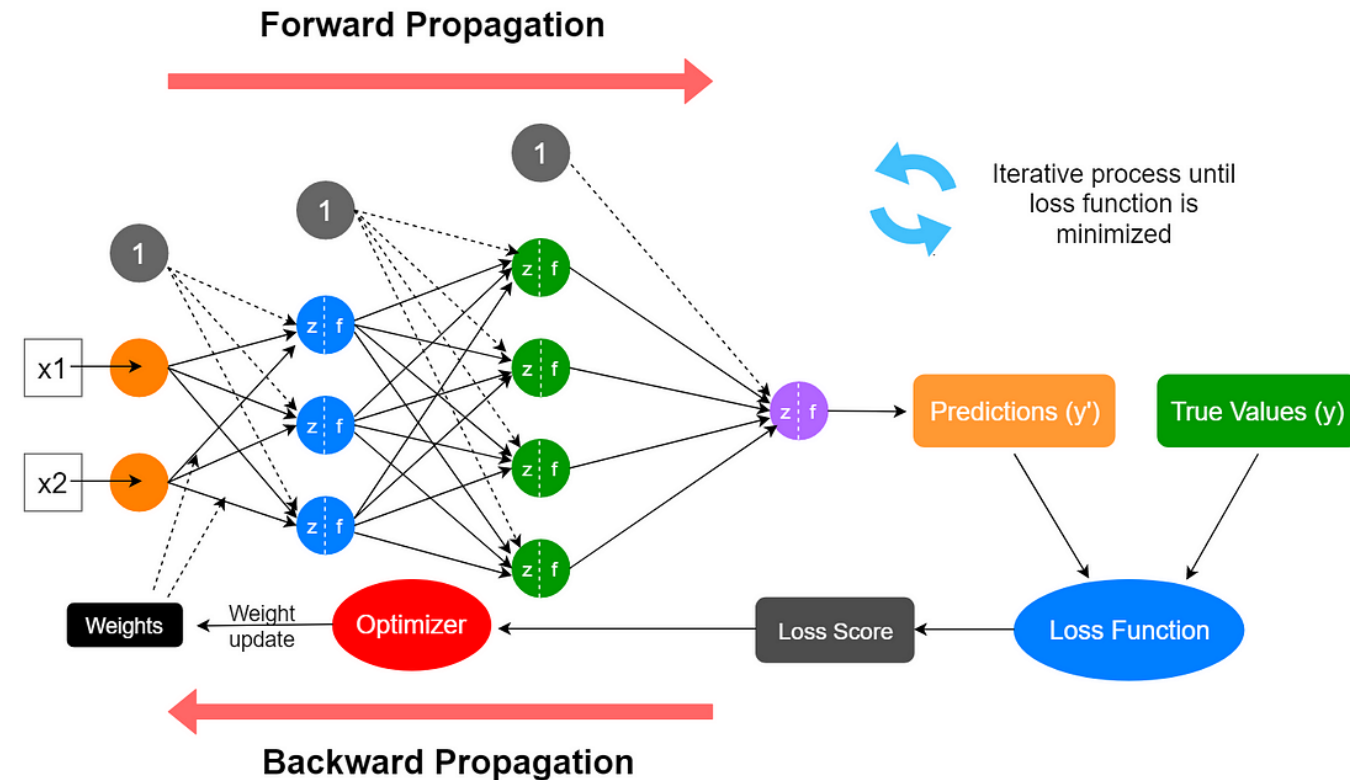
This process is known as Feedforward Network, as the information always moves in one direction (forward).

If the information moves backward, this is called Backpropagation, sometimes abbreviated as "backprop". Backpropagation acts as a messenger, informing the Neural Network whether it made a mistake when making a prediction.

Backpropagation goes through the following steps:

- The Neural Network guesses about the data
- The Neural Network is measured with a loss function, also called a cost function.
- If an error occurs, it will be backpropagated to be adjusted and corrected.

If the network makes a wrong guess about the data, Backpropagation will take this error and adjust the neural network's parameters to reduce the number of errors.

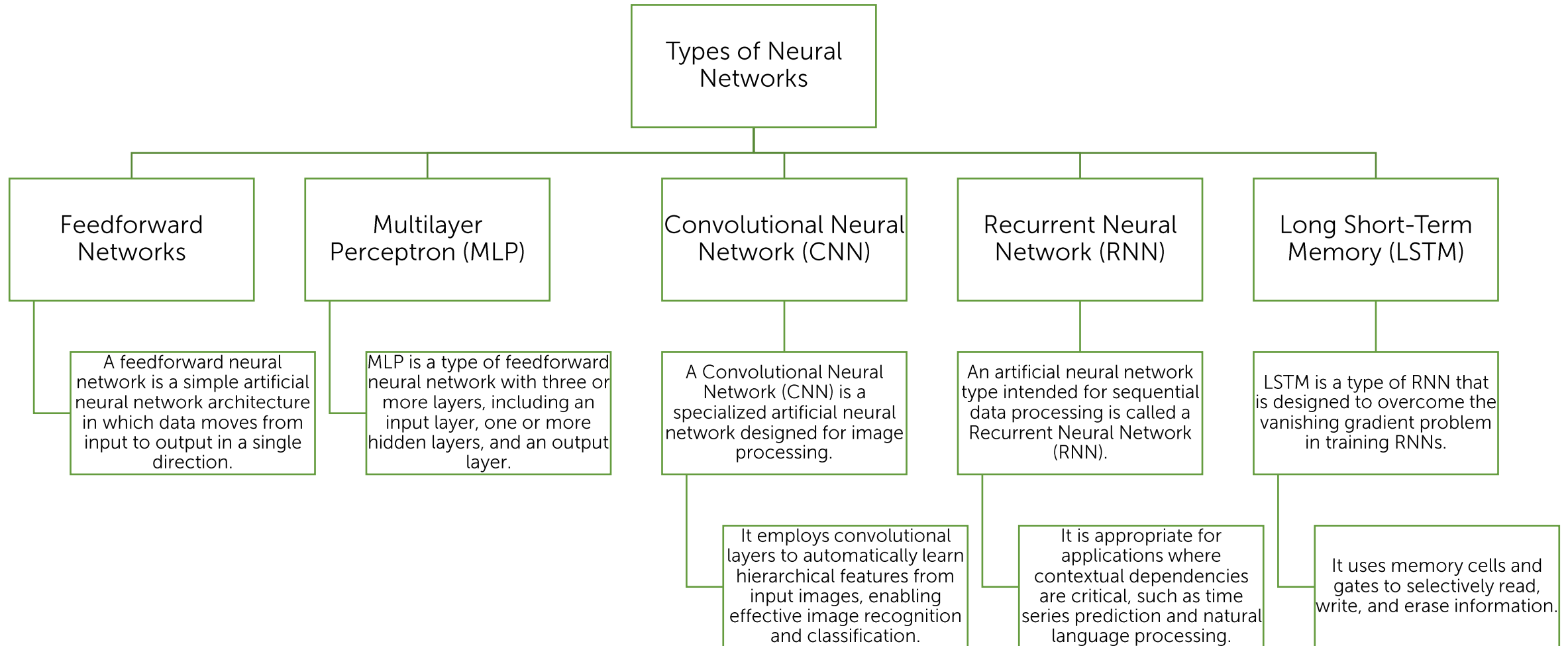


Source: <https://doi.org/10.1007/978-3-319-73004-2>

18.2 Introduction to DL

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Types of Neural Networks



Feed Forward Neural Networks (FNNs)

Definition

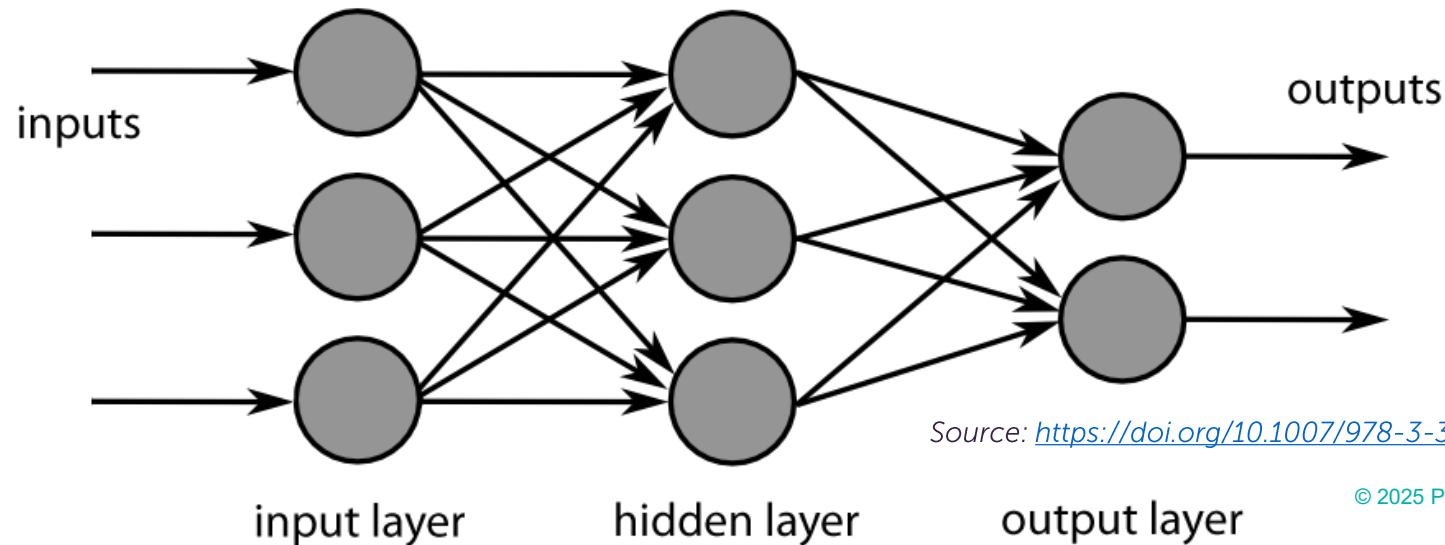
- FNNs are a form of ANN in which inputs can pass data through those nodes within the hidden level to the output nodes without forming any cycles between layers or nodes.

Architecture

- Made up of layers with unidirectional data flow (from input through hidden and output layers).

Applications

- In visual and voice recognition, NLP, financial forecasting, and recommending systems.



Source: <https://doi.org/10.1007/978-3-319-73004-2>

Recurrent Neural Networks (RNN)

Definition

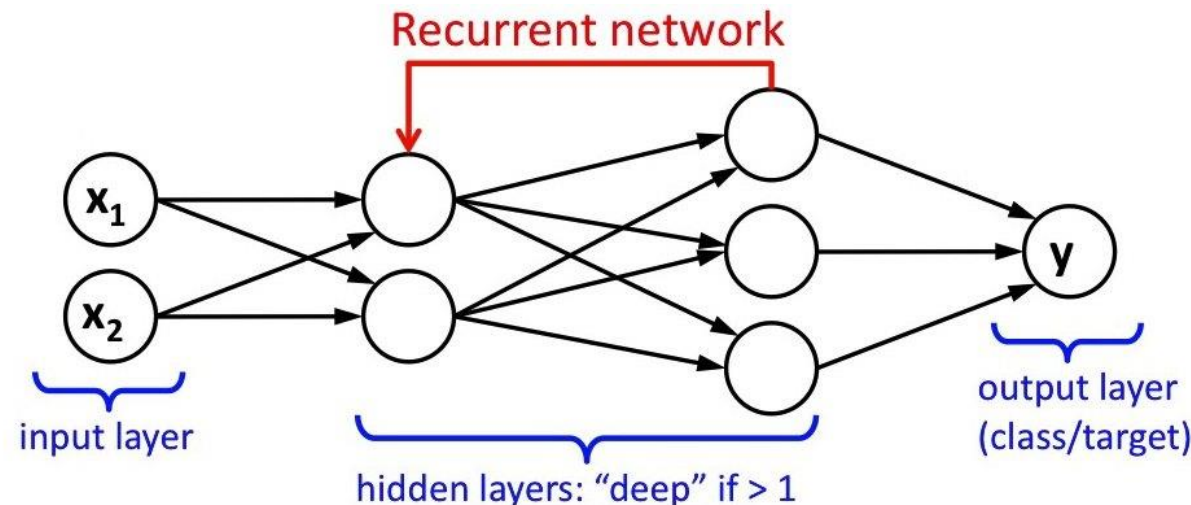
- Recurrent neural network handles sequential data in which the current output is a result of previous inputs by looping over themselves to hold internal state (memory).

Architecture

- Contains recurrent connections that enable feedback loops for processing sequences.

Applications

- Language translation, open-ended text classification, ones to ones interaction, and time series prediction are its applications.



Convolutional Neural Network (CNN)

Definition

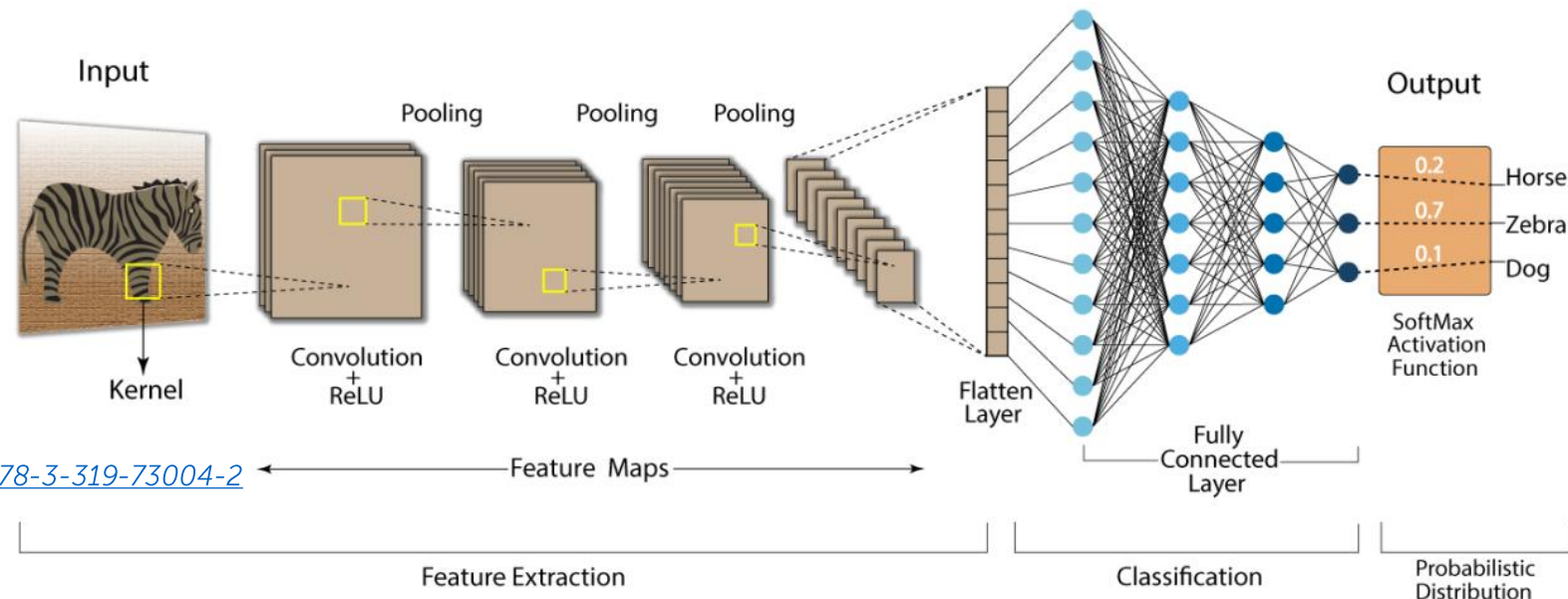
- The convolutional neural network's structure is focused on processing the grid type data like images and videos by using convolutional layers filtering driving the patterns and spatial hierarchies.

Architecture

- Convolutional Neural Network consists of multiple layers like the input layer, Convolutional layer, Pooling layer, and fully connected layers.

Applications

- Used for classification of images, object detection, medical imaging analyzes, autonomous driving and visualization in augmented reality.



Source: <https://doi.org/10.1007/978-3-319-73004-2>

Long Short-Term Memory (LSTM)

Definition

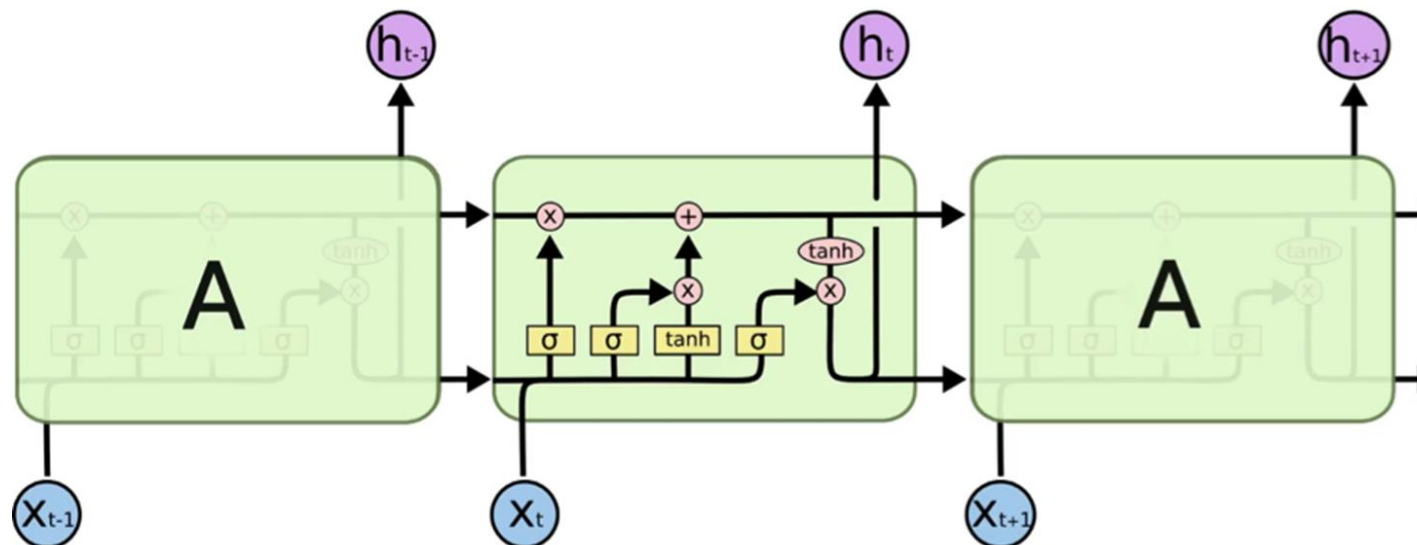
- LSTM networks, as a recurrent neural network variant, exhibit memory cells to solve the disappearing gradient issue and keep large ranges of information in their memory.

Architecture

- Capture memory cells in pass information flowing and graduate greediness issue.

Applications

- Value of RNNs is in terms of importing long-term memory into the model e.g., language translation, and time-series forecasting.



18.2 Introduction to DL

- Artificial Neural Networks (ANNs)
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- How does Deep Learning work?
- Types of Neural Networks
- **Advantages and Disadvantages of DL**

Advantages and Disadvantages of DL

Advantages	Disadvantages
Automatic Feature Learning: DL algorithms automatically learn features from data, useful for tasks like image recognition.	High Computational Cost: Training DL models requires significant resources, including GPUs and memory, which can be costly.
Handling Large and Complex Data: DL can process massive datasets, making it effective for big data analysis.	Overfitting: Deep models can overfit the training data, leading to poor performance on new data.
Improved Performance: DL has shown superior results in tasks like image, speech recognition, and NLP.	Lack of Interpretability: Deep models are complex, making it difficult to understand or explain predictions.
Non-linear Relationships: DL can identify complex, non-linear patterns in data.	Dependence on Data Quality: DL performance heavily depends on the quality and cleanliness of the training data.
Handling Structured and Unstructured Data: DL can process both structured data (tables) and unstructured data (images, text, audio).	Data Privacy and Security: DL requires large amounts of data, raising concerns about privacy and security.

Advantages and Disadvantages of DL

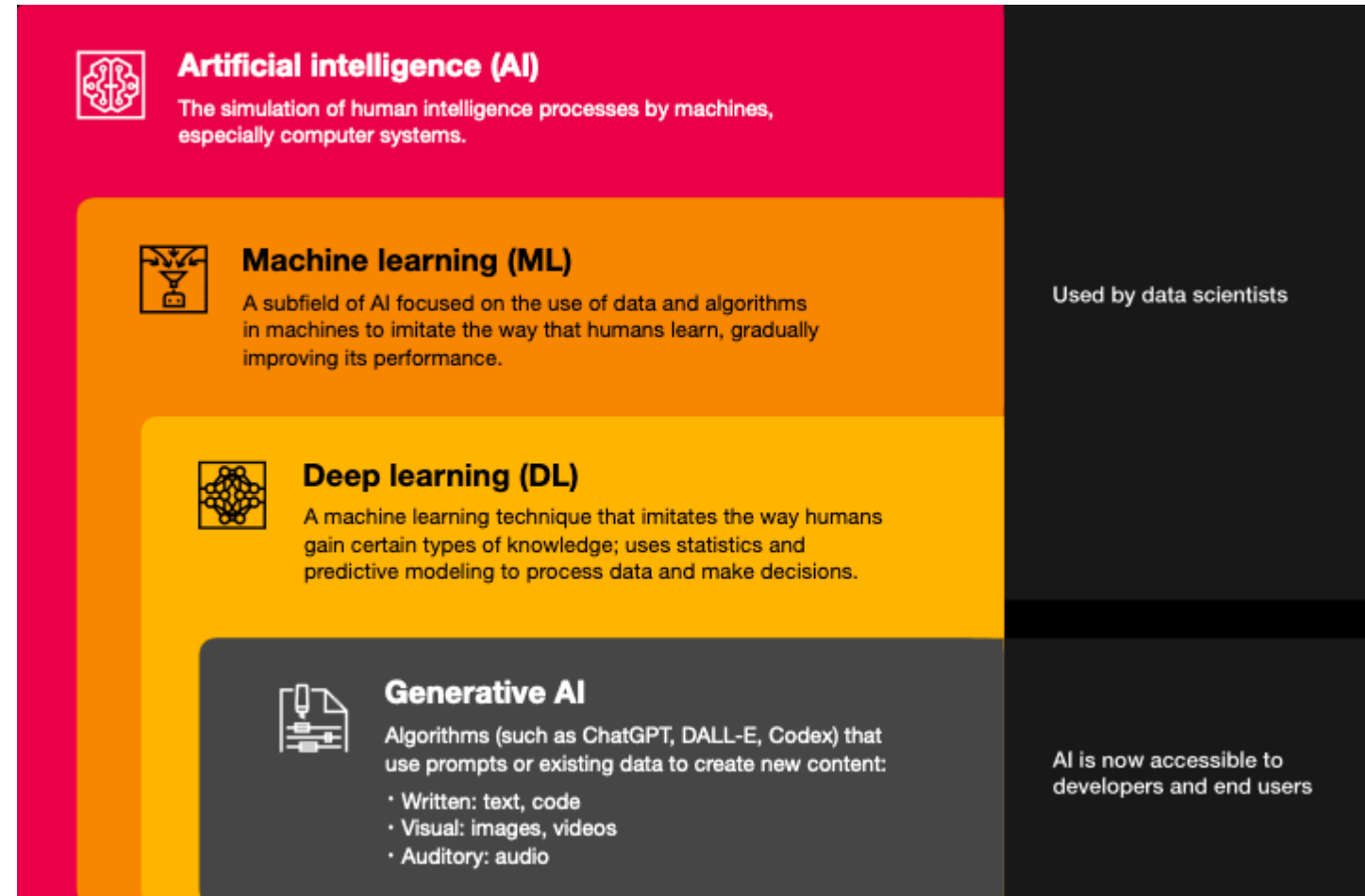
Advantages	Disadvantages
Predictive Modeling: DL models can predict future trends, aiding in decision-making and planning.	Lack of Domain Expertise: Effective DL implementation requires domain knowledge to formulate the problem and choose the right algorithm.
Handling Missing Data: DL can still make predictions even when some data is missing.	Unforeseen Consequences: Biases or ethical issues can arise, leading to unintended harmful consequences.
Handling Sequential Data: RNNs and LSTMs are suited for time-series, speech, and text with memory over time.	Limited to Training Data: DL models can only make predictions based on the data they were trained on and may not generalize well to new contexts.
Scalability: DL models scale well to handle increasing data, and can be deployed on cloud or edge devices.	Black Box Models: Some DL models are "black boxes," making it hard to interpret how decisions are made.
Generalization: DL models can generalize well to new, unseen situations by learning abstract data representations.	

18.3 Introduction to Gen AI

- Evolution of Gen AI
- Types of Gen AI Models
- Applications of Gen AI Models
- Benefits of Gen AI
- Limitations of Gen AI

Introduction to Gen AI

- Generative Artificial Intelligence, often called Generative AI or Gen AI, is a type of AI or subset of DL that can create new content like conversations, stories, images, videos, and music.
- It can learn about different topics such as languages, programming, art, science, and more, and use this knowledge to solve new problems.
- For example: It can learn about popular design styles and create a unique logo for a brand or an organisation.
- Businesses can use generative AI in many ways, like building chatbots, creating media, designing products, and coming up with new ideas.

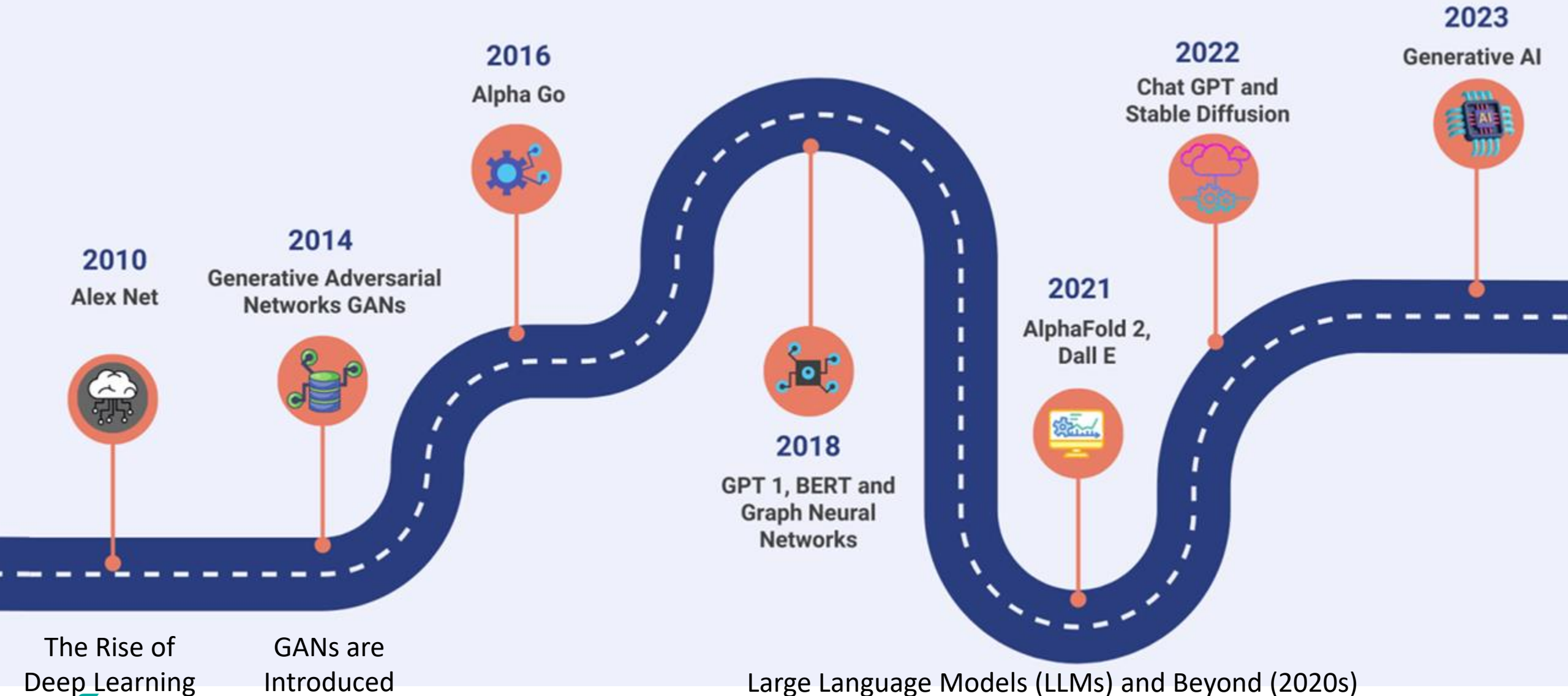


Source: [PWC](#)

18.3 Introduction to Gen AI

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Evolution of Gen AI



Evolution of Gen AI

The Rise of Deep Learning (2010s)

- Deep learning improved AI significantly by using neural networks, which mimic how the human brain works. AI could now process much more complex data, like thousands of photos, and start generating new content.
- Example: AI could now create a realistic drawing of a dog by learning from millions of dog photos.

Generative Adversarial Networks (2014)

- GANs, introduced in 2014, use two AI systems that work together: one generates new content, and the other checks if it looks real. This made generative AI much better at creating realistic images, videos, and sounds.
- Example: GANs can create life like images of people who don't exist or filters (used in apps like FaceApp or Snapchat).

Evolution of Gen AI

Large Language Models (LLMs) and Beyond (2020s)

- Models like GPT-3 and GPT-4 can understand and generate human-like text. They are trained on massive amounts of data from books, websites, and other sources. AI can now hold conversations, write essays, generate code, and much more.
- Example: ChatGPT can help you draft an email, write a poem, or even solve problems.

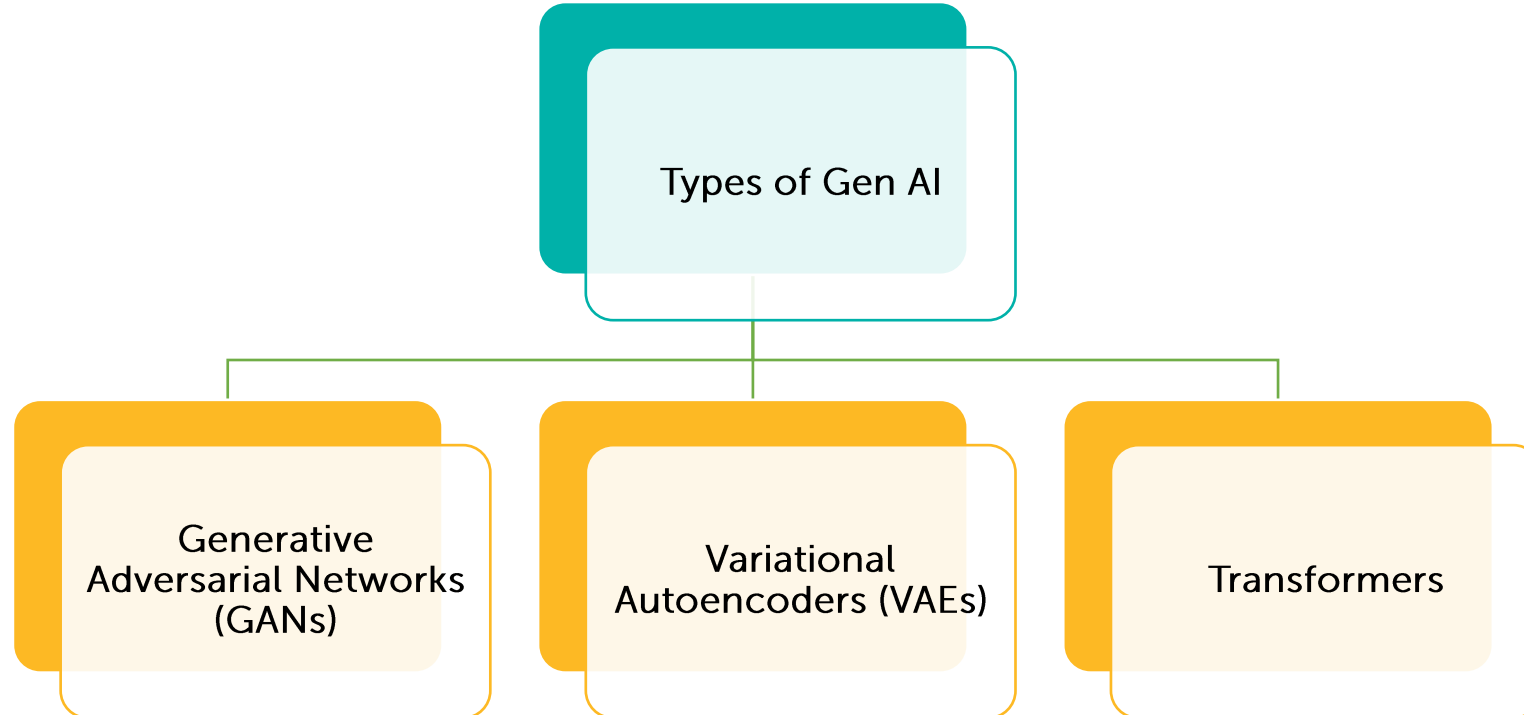
Multimodal Generative AI (Present)

- New AI models can handle multiple types of data at once—text, images, audio, and video. This allows AI to create content that combines different formats.
- Example: AI can take a written description and turn it into an animated video or a song with the help of different models integrating together.

18.3 Introduction to Gen AI

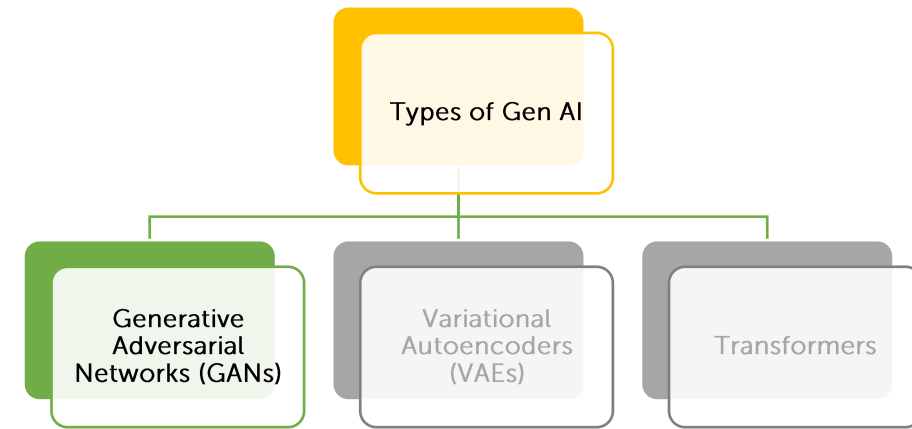
- Evolution of Gen AI
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Types of Gen AI Models



Types of Gen AI Models

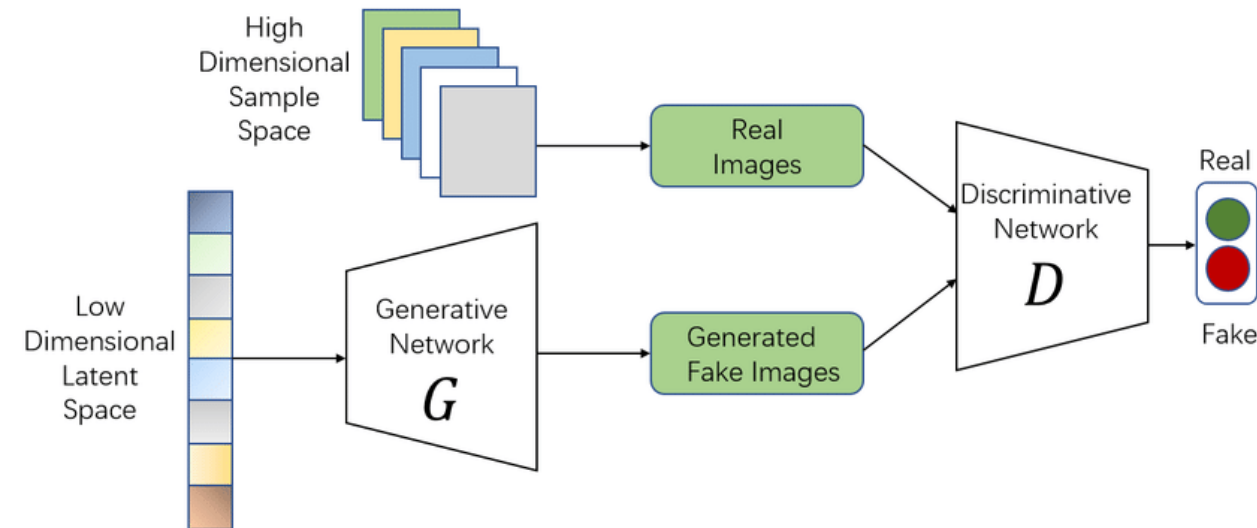
Generative Adversarial Networks (GANs)



Generative Adversarial Network (GAN) architecture consisting of two essential components: a **Generator Network** and a **Discriminator Network**.

The generator's primary function is to generate synthetic data that closely resembles real data, while the discriminator is responsible for distinguishing between authentic and fabricated data.

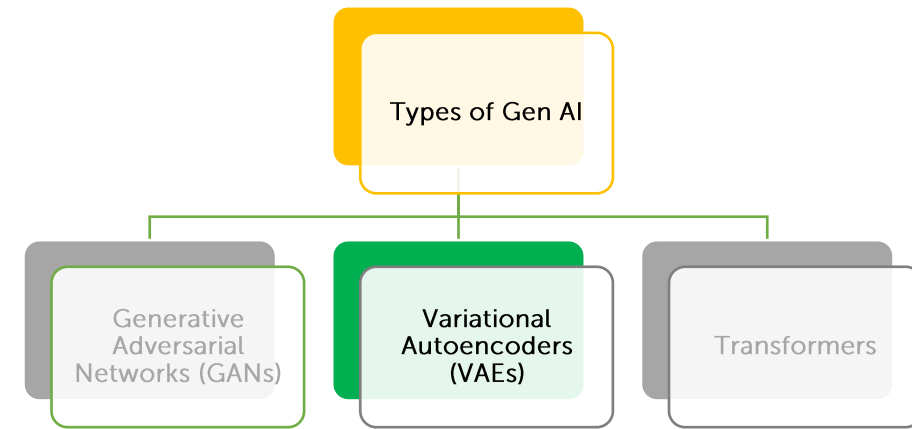
The generator enhances the authenticity of its produced data through adversarial training, while the discriminator effectively determines whether the data is real or synthetic.



Source: <https://developers.google.com/machine-learning/>

Types of Gen AI Models

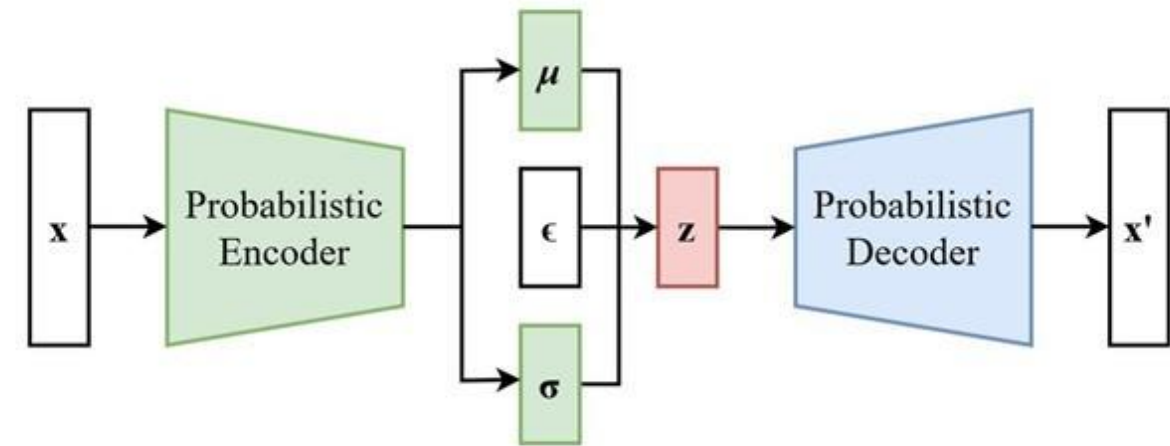
Variational Autoencoders (VAEs)



VAEs combine the capabilities of autoencoders and probabilistic modeling to acquire a compressed representation of data.

VAEs encode input data into a lower-dimensional latent space, allowing the generation of new samples by sampling points from the acquired distribution.

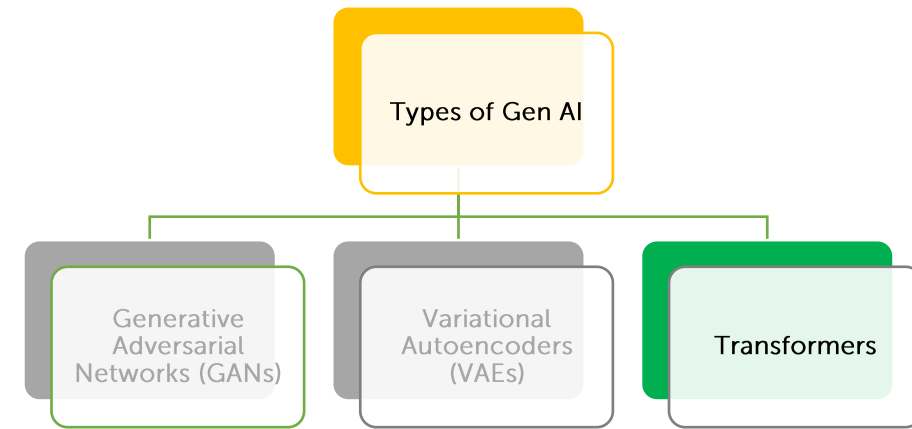
With practical applications spanning image generation, data compression, anomaly detection, and drug discovery, VAEs exhibit versatility across various domains.



Source: <https://developers.google.com/machine-learning/>

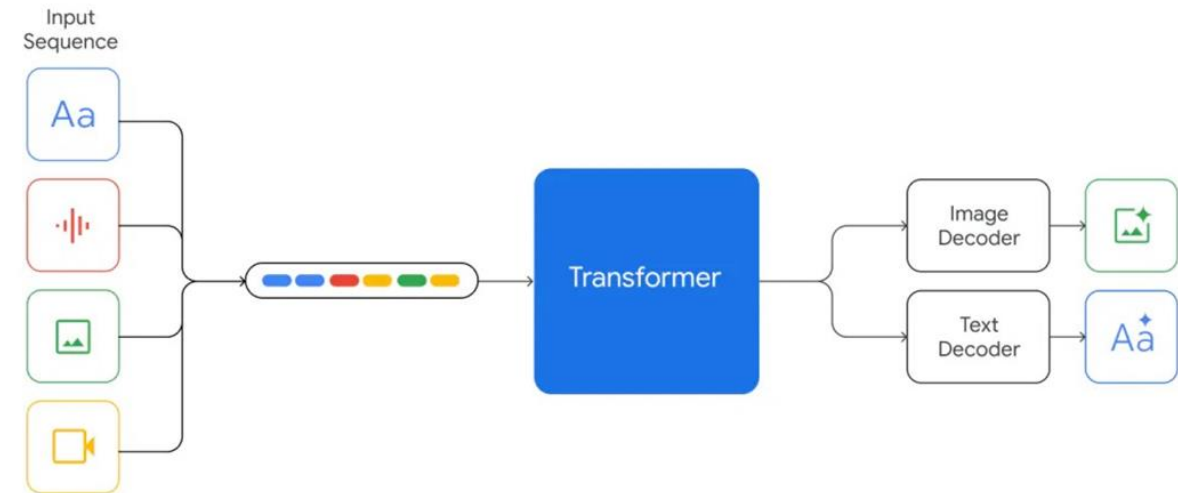
Types of Gen AI Models

Transformers



Multiple projects like Google's BERT and OpenAI's GPT-3 and GPT-4, two of the most powerful generative AI models, are based on the Transformer architecture.

These models can be used to generate human-like text, help with coding tasks, translate from one language to the next, and even answer questions on almost any topic.

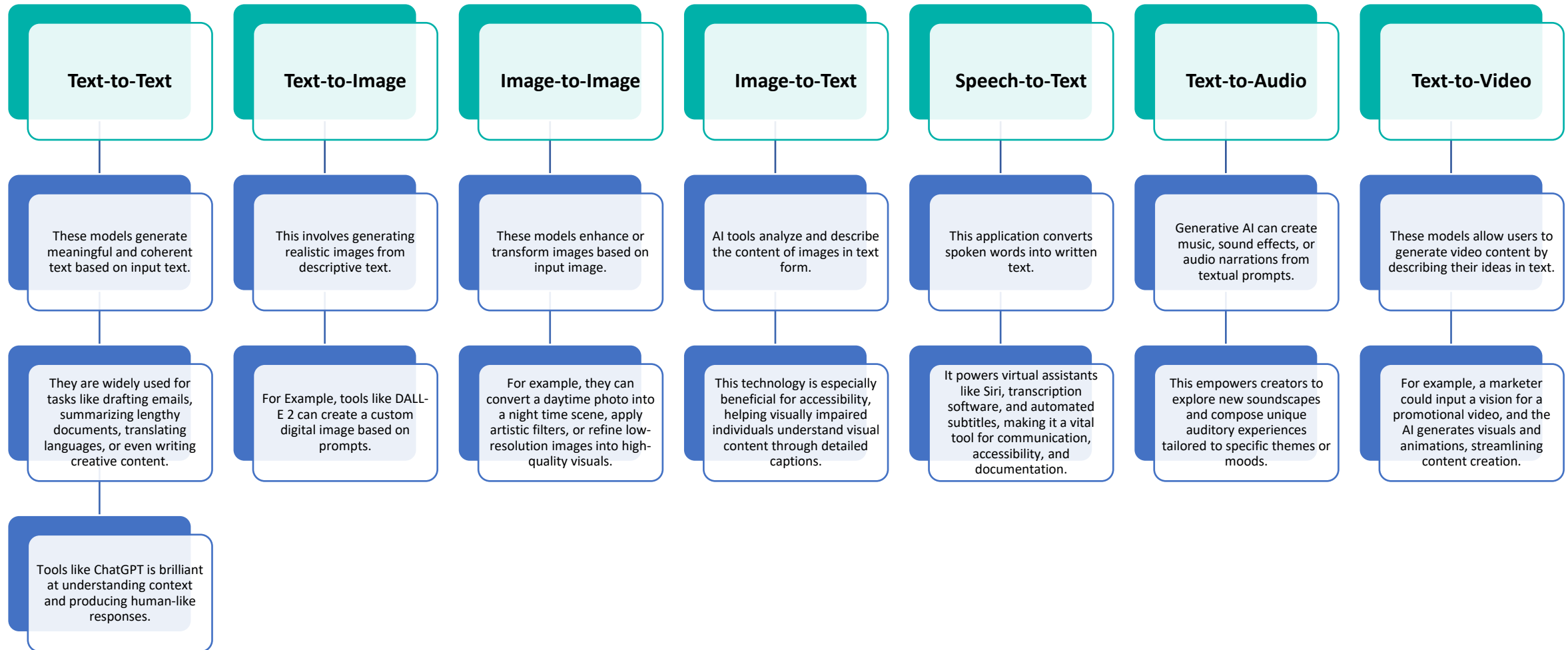


Source: <https://developers.google.com/machine-learning/>

18.3 Introduction to Gen AI

- Evolution of Gen AI
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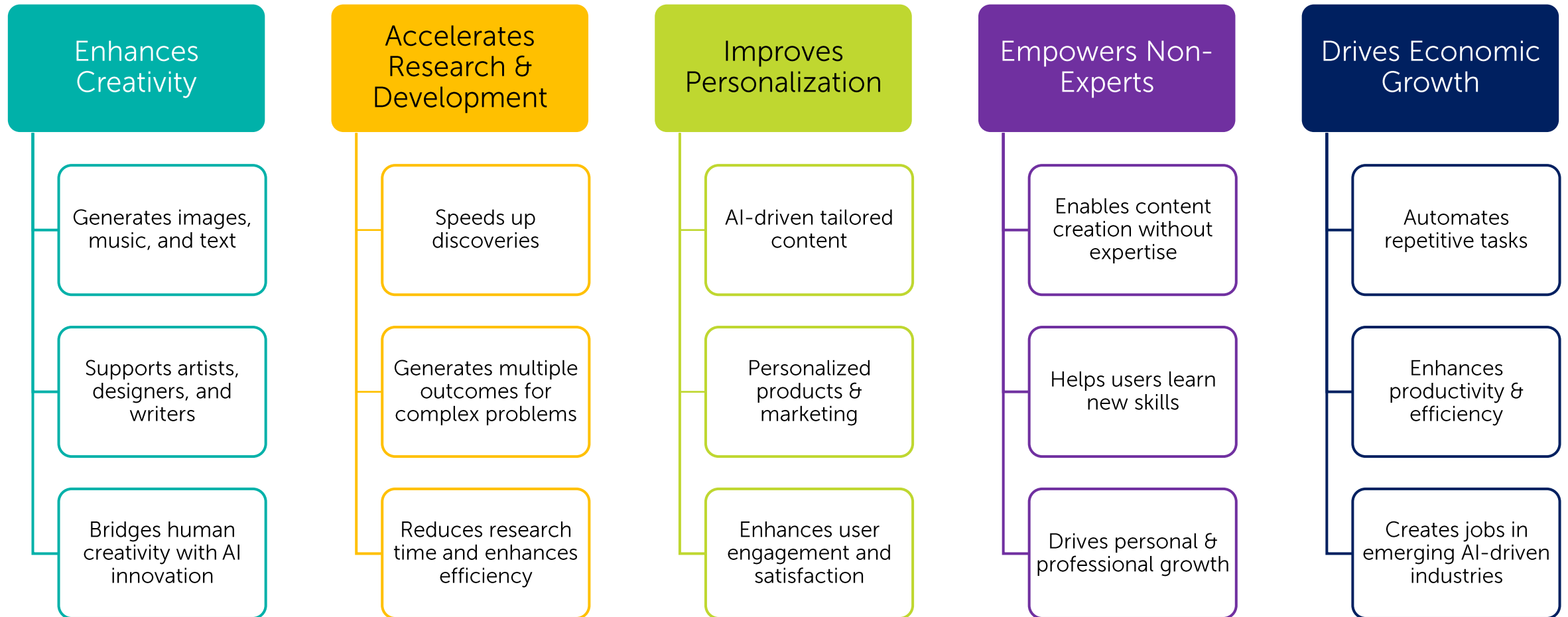
Applications of Gen AI



18.3 Introduction to Gen AI

- Evolution of Gen AI
- Types of Gen AI Models
- Applications of Gen AI
- **Benefits of Gen AI**
- Limitations of Gen AI

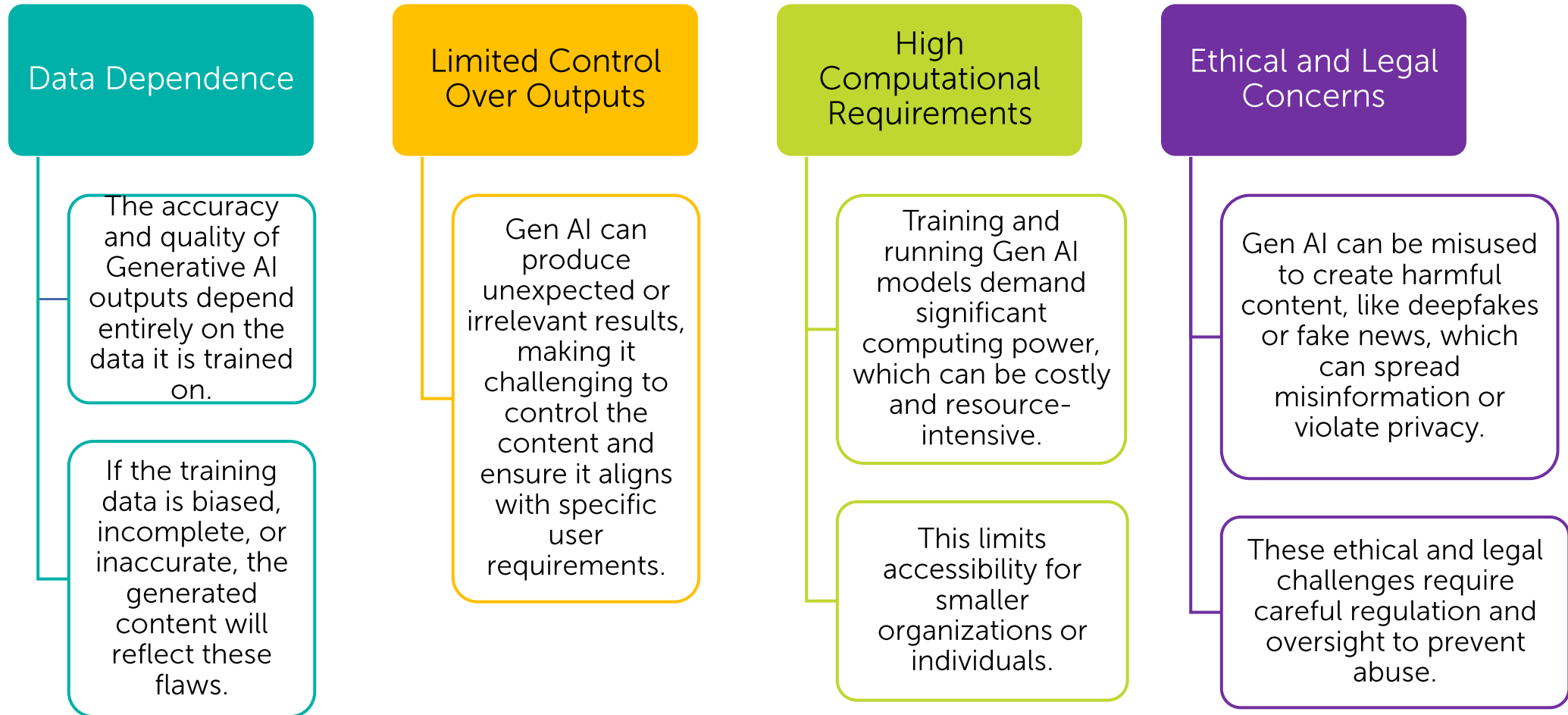
Benefits of Gen AI



18.3 Introduction to Gen AI

- Evolution of Gen AI
- Types of Gen AI Models
- Applications of Gen AI
- Benefits of Gen AI
- **Limitations of Gen AI**

Limitations of Gen AI



18.4 ML vs DL

- Comparison between ML and DL
- Challenges of ML and DL

18.4 ML vs DL

- Comparison between ML and DL
- Challenges of ML and DL

ML vs DL

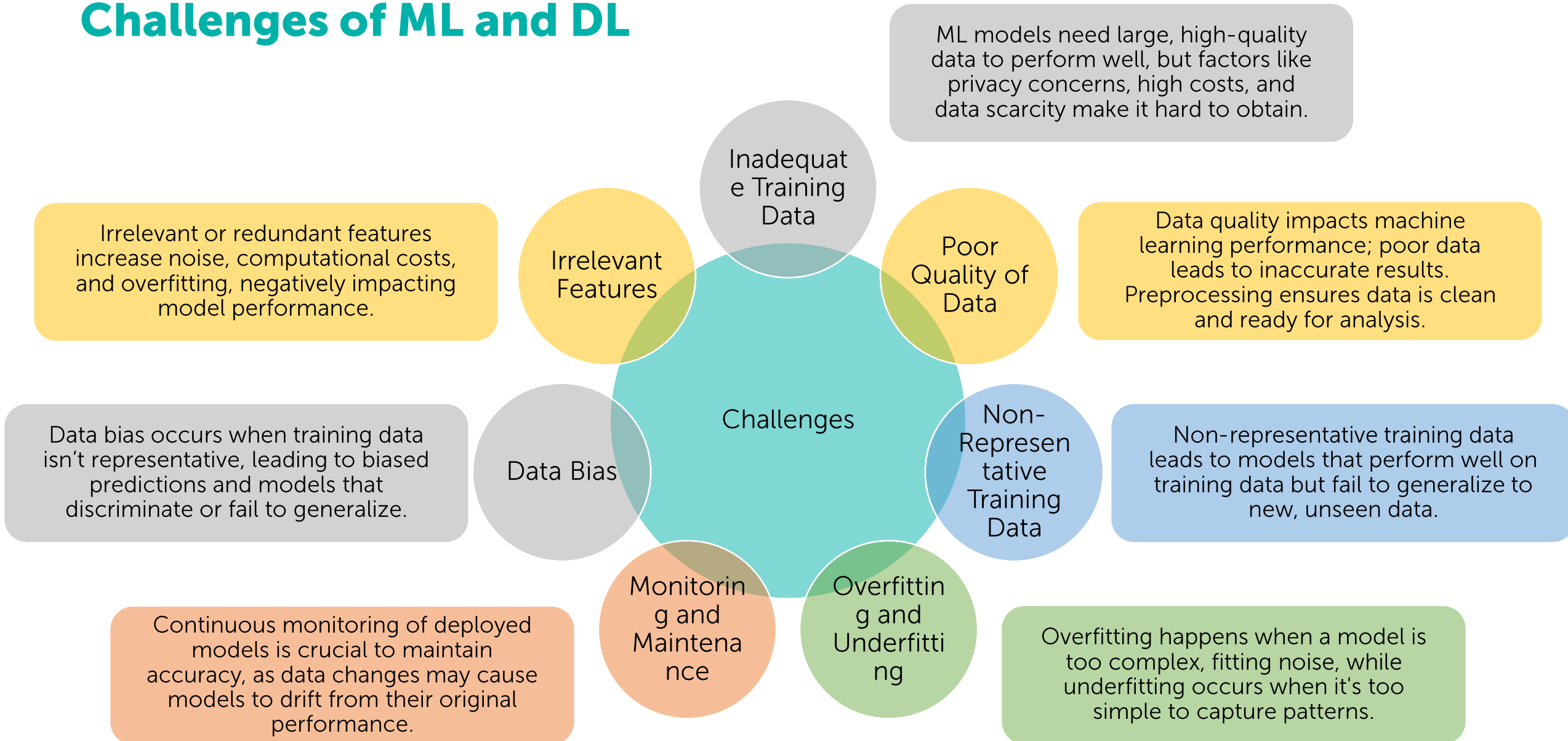
Comparison between ML and DL

Machine Learning	Deep Learning
Apply statistical algorithms to learn the hidden patterns and relationships in the dataset.	Uses artificial neural network architecture to learn the hidden patterns and relationships in the dataset.
Can work on the smaller amount of dataset	Requires the larger volume of dataset compared to machine learning
Better for the low-label task.	Better for complex task like image processing, natural language processing, etc.
Takes less time to train the model.	Takes more time to train the model.
A model is created by relevant features which are manually extracted from images to detect an object in the image.	Relevant features are automatically extracted from images. It is an end-to-end learning process.
Less complex and easy to interpret the result.	More complex, it works like the black box interpretations of the result are not easy.
It can work on the CPU or requires less computing power as compared to deep learning.	It requires a high-performance computer with GPU.

18.4 ML vs DL

- Comparison between ML and DL
- Challenges of ML and DL

Challenges of ML and DL



18.5 Role of Metaheuristic Algorithms in AI

Role of Metaheuristic Algorithms in AI

Metaheuristic algorithms play a significant role in Artificial Intelligence (AI) by providing efficient solutions to complex optimization problems that are difficult to solve using traditional methods.

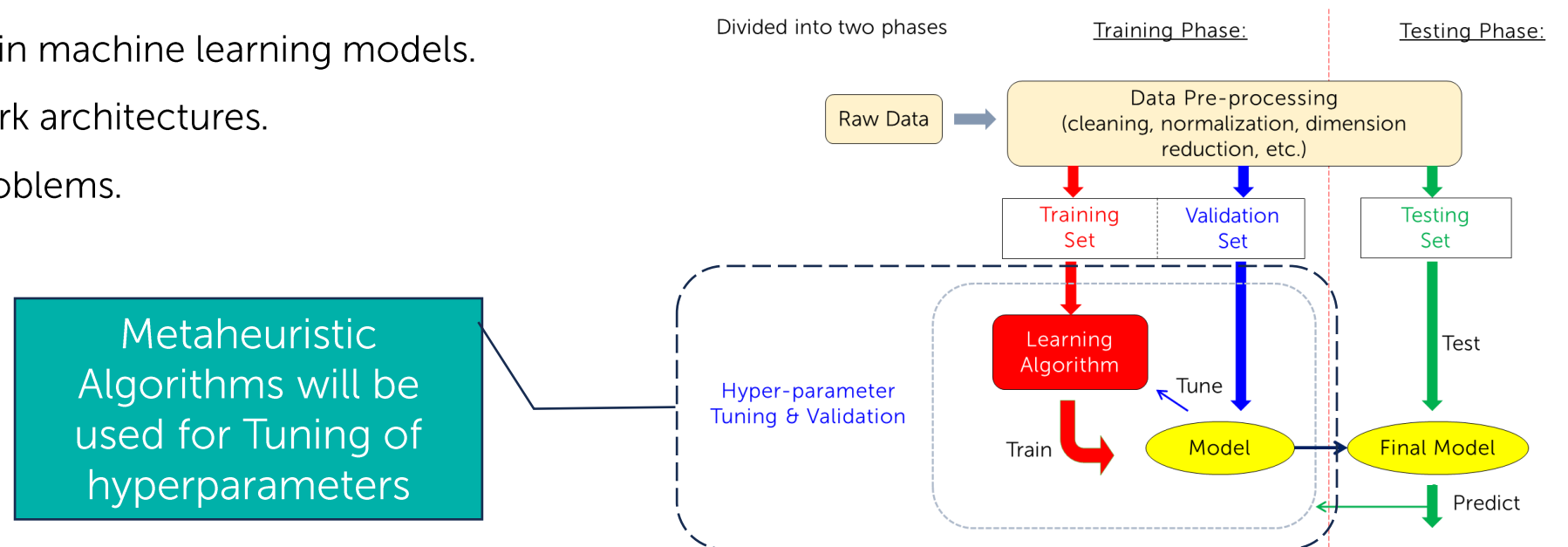
Metaheuristic algorithms are primarily used for optimization tasks in AI.

Finding the best solution from a large set of possible solutions.

Examples:

- Tuning hyperparameters in machine learning models.
- Optimizing neural network architectures.
- Solving combinatorial problems.

Machine Learning Framework

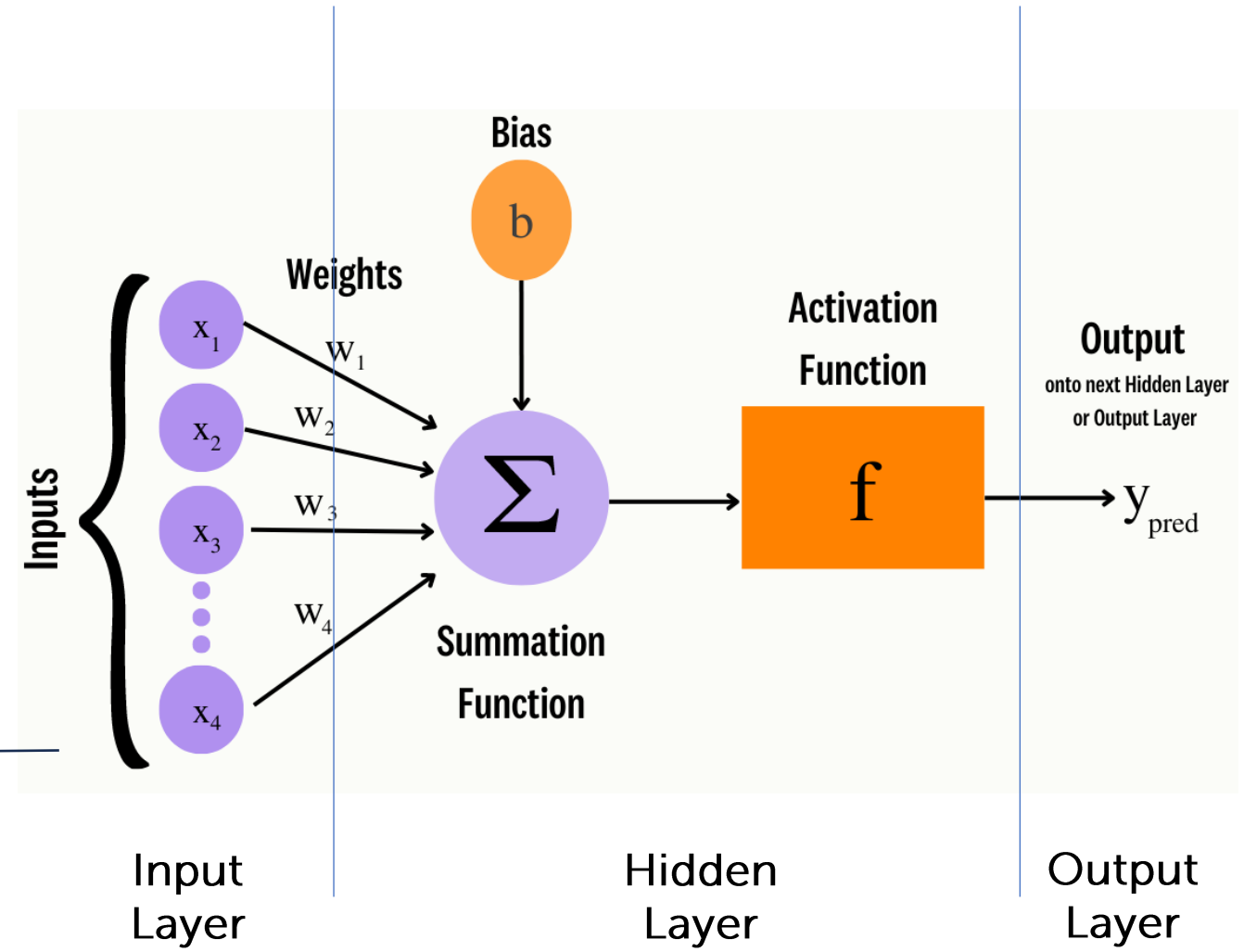


Role of Metaheuristic Algorithms in AI

Applications in AI

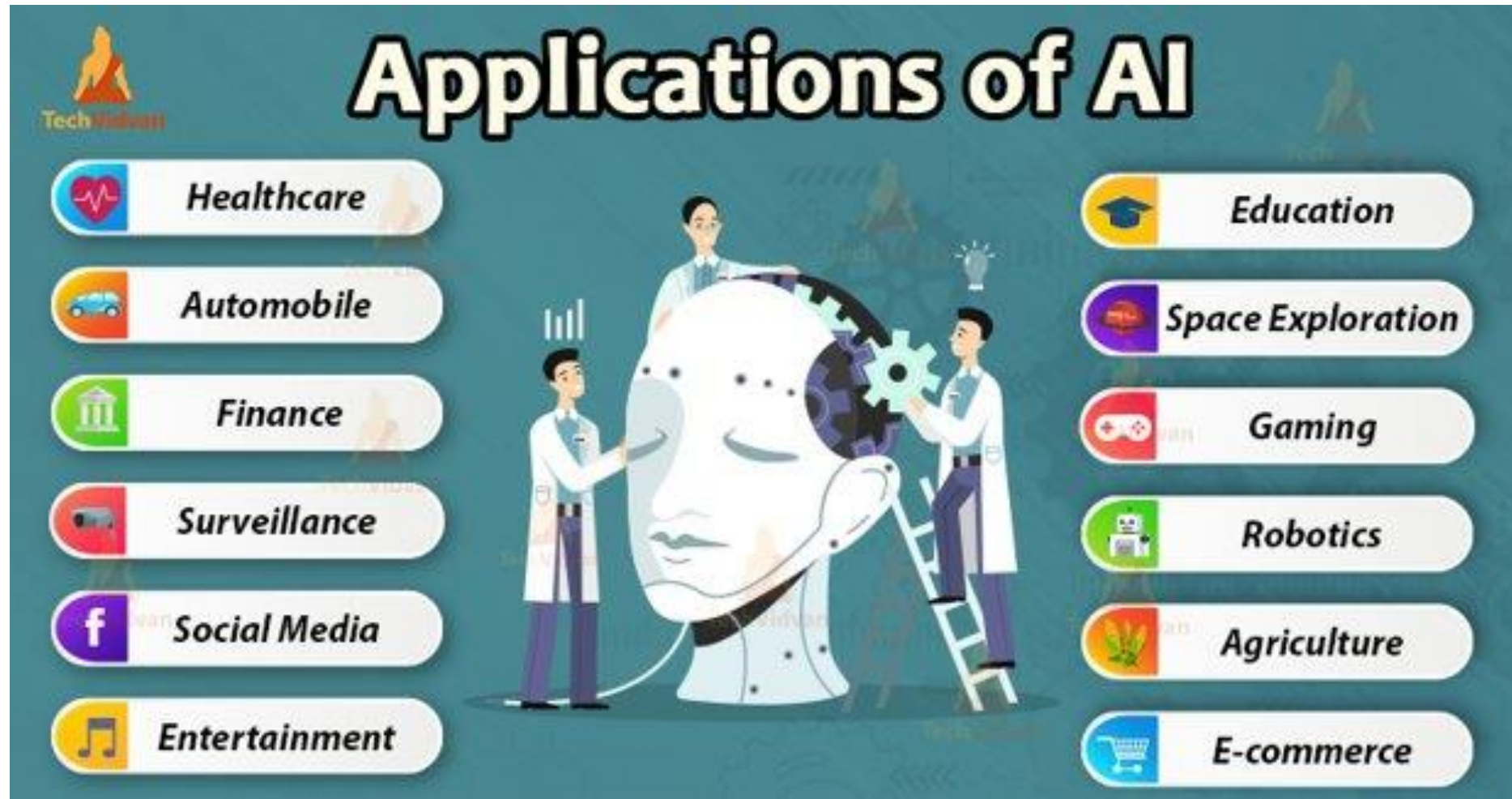
- Machine Learning
 - Feature selection, hyperparameter tuning, optimizing model parameters.
- Neural Networks
 - Optimizing architecture and weights.

Metaheuristic Algorithms will be used for finding optimal architecture and weights



18.6 Applications of AI

Applications of AI

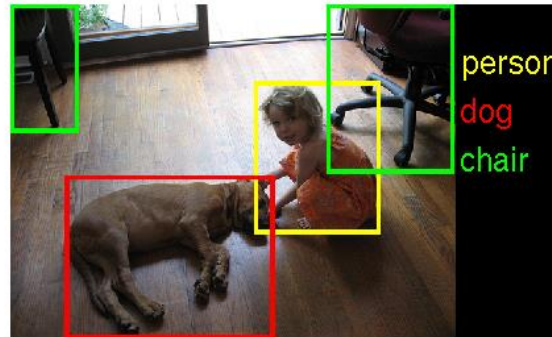


Source: <https://doi.org/10.1007/s12178-020-09600-8>

Applications of AI – Computer Vision

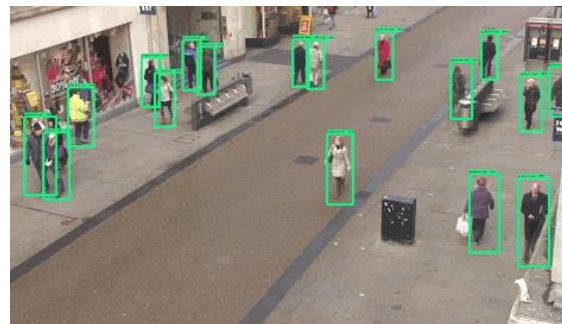
Visual Object Detection

- Input data: images
- Output: presence/absence of particular objects in the image



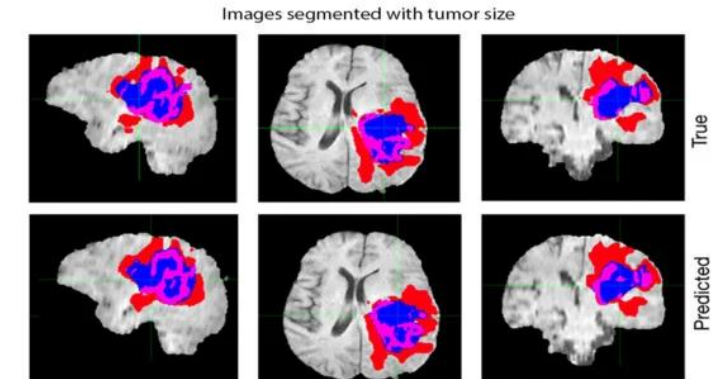
Video Surveillance

- Input data: video
- Output:
 - Human activity detection
 - Abnormal event detection



Medical Imaging

- Input data: images
- Output: presence/absence of Tumor in the image

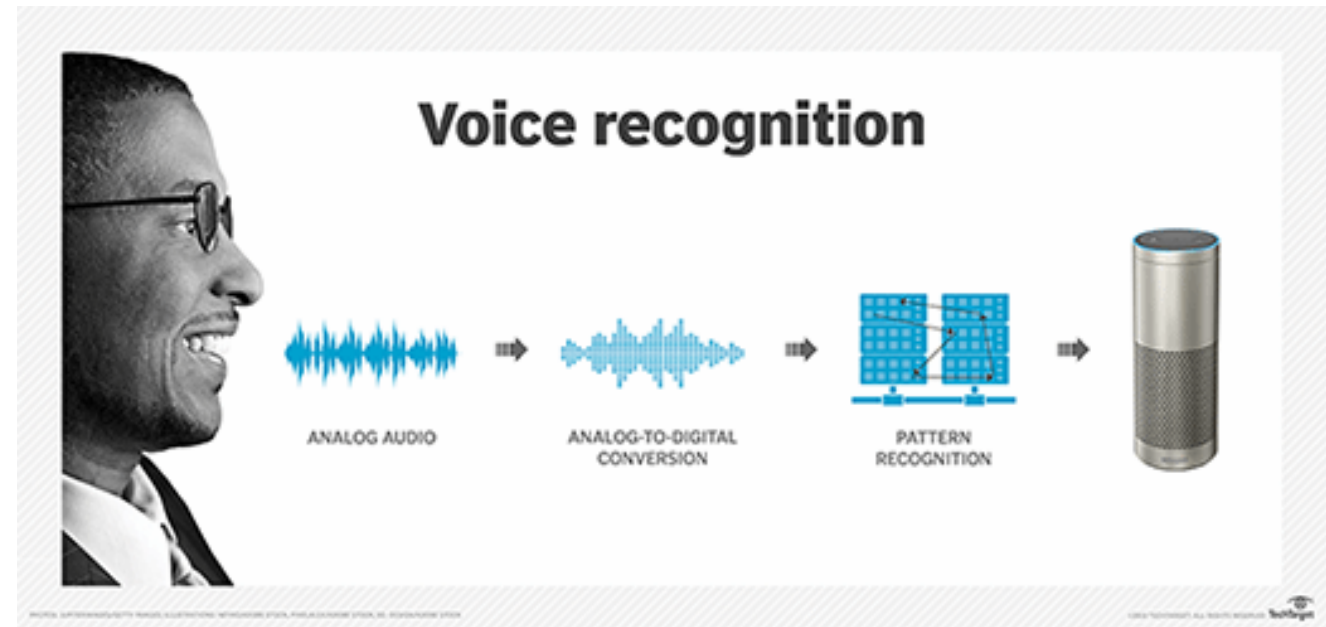


Source: <https://doi.org/10.1007/s12178-020-09600-8>

Applications of AI – Voice Recognition

Speech Technologies (e.g., Siri, Alexa)

- Input data: audio waveform
- Output:
 - Text characters
 - Automatic speech recognition (ASR)

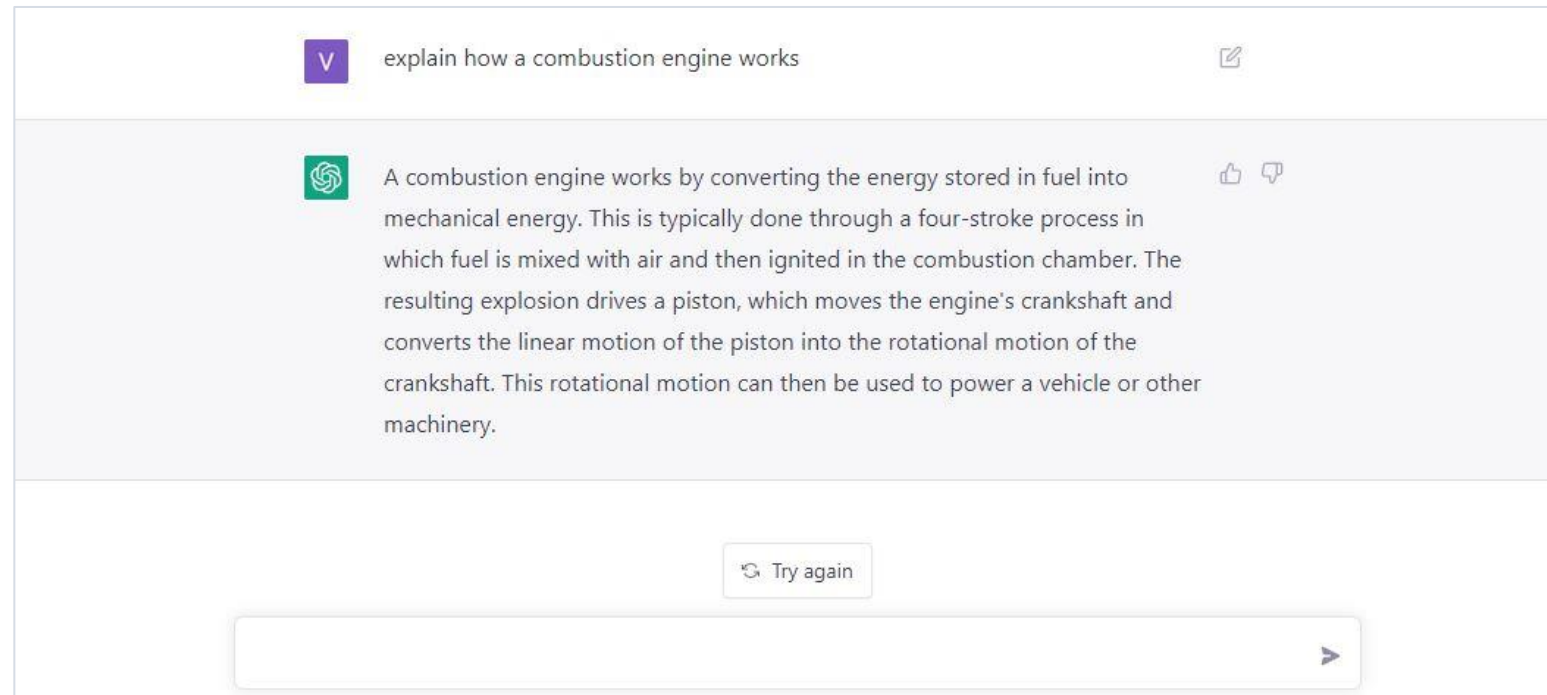


Source: <https://doi.org/10.1007/s12178-020-09600-8>

Advantages of AI - Natural Language Processing

Language Processing Technologies

- Input data: text
- Output:
 - Machine translation
 - Question answering, personal assistant
 - Text classification, spam filtering, etc.



Source: <https://chatgpt.com/>

Applications of AI - More!

Manufacturing



Healthcare

Games



Route Planning

Source: <https://doi.org/10.1007/s12178-020-09600-8>

